AC – 15/6/2026
Item No. – 2.12(N)

As Per NEP 2020

TERNA ENGINEERING COLLEGE

Bachelor of Engineering Computer Engineering
Second Year (S.E.) Syllabus

Semester III to IV
for
Academic Year 2026-27

Prepared by: Board of Studies: Date: 14/06/2026
for
Computer Engineering
Approved By: Academic Council: Date:15/06/2026
of
Terna Engineering College

Rev-2026

Ref: As per GR dated 4th July 2023 for Credit Structure of UG
(As per AICTE & NEP 2020 Guidelines)

PREAMBLE

"Think. Design. Innovate. – Learning that Leads the Future."

Terna Engineering College is proud to be among the oldest institutions to attain academic autonomy. This achievement strengthens our commitment to becoming a world-class center for socioeconomic development and quality education. It is with great pride and purpose that we introduce the restructured curriculum under academic autonomy at **Terna Engineering College**. This initiative marks a significant step toward transforming engineering education by equipping our graduates with strong technical foundations, research acumen, interdisciplinary perspective, and an innovation-driven
Mindset is essential for success in today's evolving professional landscape.

1. **Curriculum Design:** Theme-Based, Experiential, and Outcome-Oriented. The autonomous curriculum adopts a bottom-up, learner-centric approach aligned with all 11 Program Outcomes. Each semester is driven by a central societal or technological theme, with problem statements rooted in local needs, socio-economic priorities, and the **UN Sustainable Development Goals (UNSDGs)**. Students engage with these challenges through individual projects in the **NIRMAN LAB,** integrating social science methodologies with engineering solutions to foster innovation, contextual learning, and socially responsible problem-solving.
2. **Purpose of Autonomy:** Empowering Innovation and Real-World Learning. Our pursuit of autonomy is rooted in bridging the gap between academic learning and industry relevance. By integrating undergraduate research through the **NIRMAN LAB**—our dedicated innovation and problem-solving hub—students engage with real-world challenges, apply interdisciplinary knowledge, and develop socially impactful solutions. This approach nurtures engineers who are not only technically sound but also responsible and creative thinkers.
3. **Alignment with National Education Policy-2020:** Aligned with NEP 2020, the curriculum emphasizes multidisciplinary, hands-on learning. Through theme-based projects, electives, value-added courses, and flexible assessments, **Terna Engineering College** ensures holistic student development driven by real-world application and academic rigor. **Terna Engineering College**, as a part of **TPCTs**, benefits from being in a network of HEIs. All Higher Education Institutions under **TPCTs** can collectively contribute to curriculum development, promoting multidisciplinary studies.
4. **Innovative Pedagogy Initiative by IQAC:** As part of its transformation strategy, IQAC has introduced an innovative pedagogy model centered on interdisciplinary research and contextual problem-solving. The **NIRMAN LAB** enables students to apply subject knowledge through socially relevant projects, supported by design thinking and faculty mentorship. A two-tier assessment framework—covering both project execution and subject-wise application—ensures academic depth, continuous mentoring, and quality assurance.
5. **Empowering Faculty as Mentors and Innovators:** Terna Engineering College believes that the success of academic autonomy lies in the hands of its faculty. By empowering educators to think beyond conventional boundaries, we enable them to instill Knowledge, Skills, and Attitudes (KSA) in students while enriching their own professional growth. Faculty serve as research mentors and project guides, supported through structured training in innovation pedagogy and assessment.

Sincerely

Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-Academic Council

Dr. Deven Shah
Principal
Chairman-Academic Council

PREFACE

The Department of Computer Engineering is pleased to present the curriculum for S.E. Semester III and Semester IV under the autonomous framework aligned with the principles of the National Education Policy (NEP) 2020. The curriculum has been carefully designed to promote outcome-based education, multidisciplinary learning, innovation, entrepreneurship, and industry relevance while addressing societal needs.

The curriculum for these semesters is developed around two thematic pillars that provide students with opportunities to apply computing knowledge to real-world challenges and contribute meaningfully to national development.

Semester III Theme: "Arogya Pranali: Intelligent Health Systems for Bharat"

This theme focuses on leveraging computing technologies to improve healthcare accessibility, affordability, and efficiency. Through courses in programming, data management, operating systems, and emerging technologies, students are encouraged to explore solutions for healthcare monitoring, telemedicine, health analytics, and intelligent healthcare systems. The theme aligns with India's vision of digital healthcare transformation and promotes the development of technology-driven solutions for societal well-being.

Semester IV Theme: "Arambh: From Prototype to Product"

This theme aims to cultivate innovation, design thinking, and entrepreneurial mind set among students. Learners are encouraged to identify local and societal challenges, develop prototypes, and transform ideas into viable products and solutions. The curriculum integrates technical knowledge with problem-solving, project-based learning, teamwork, and product development skills, enabling students to bridge the gap between academic learning and real-world implementation.

The syllabus incorporates experiential learning, project-based activities, skill development, and continuous assessment mechanisms as recommended by NEP 2020. Emphasis has been placed on developing critical thinking, creativity, communication, collaboration, and professional ethics while ensuring alignment with industry requirements and emerging technological trends.

The Board of Studies has actively consulted academic experts, industry professionals, alumni, and stakeholders during the curriculum design process to ensure that the program remains contemporary, relevant, and future-ready. The curriculum also contributes towards achieving the United Nations Sustainable Development Goals (UNSDGs), particularly those related to Good Health and Well-being, Quality Education, Industry, Innovation and Infrastructure, and Sustainable Communities.

We are confident that this curriculum will empower students to become competent computer engineers, innovative problem solvers, responsible citizens, and future technology leaders capable of contributing to the vision of a developed and digitally empowered India.



Dr. Kishor Sakure
BoS-Chairman –Computer Engineering



Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-Academic Council



Dr. Deven Shah
Principal
Chairman-Academic Council

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Abbreviations

AEC	Ability Enhancement Course
AECL	Ability Enhancement Course Laboratory
AU	Audit Course
BSC	Basic Science Course
BSCL	Basic Science Course Laboratory
CC	Co-Curricular Courses
CEP	Community Engagement Project
ELC	Experiential Learning Course
ESC	Engineering Science Course
ESCL	Engineering Science Course Laboratory
FP	Field Project
BESC	Basic Engineering Science Course
BESL	Basic Engineering Science Laboratory
HSSM	Humanities Social Sciences and Management
IKS	Indian Knowledge System
INTR	Internship
L	Lecture
LC	Laboratory Course
LLC	Liberal Learning Course
MDMC	Multidisciplinary Minor Course
MDML	Multidisciplinary Minor Laboratory
MJP	Major project
MP	Mini Project
NL	Notional Learning
OE	Open Elective
OJT	On Job Training
P	Practical
PCC	Program Core Course
PCL	Program Core Laboratory
PEC	Program Elective Course
PEL	Program Elective Laboratory
RP	Research Project
SBL	Skill-Based Laboratory
SEC	Skill Enhancement Course
SL	Self-Learning
T	Tutorial
VEC	Value Education Course
VSEC	Vocational and Skill Enhancement Course
VSC	Vocational Skill Course

Curriculum Structure and Examination Scheme
for
Second Year Bachelor of Engineering Computer Engineering
Terna Engineering College (With Effect from 2026-2027)

Semester-III

Course Code	Course Type	Course Description	Teaching & Learning Scheme								
			Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per Semester)					Total Credit (C) Total Hours (Notional Learning Hours)/30
			L	T	P	L	T	P	SL	Notional Learning Hour	
2023101	PCC301	Applied Statistics & Probability for Data Science	2	–	–	30	--	--	30	60	2
2023102	PCC302	Algorithms for Intelligent Problem Solving	3	--	–	45	--	--	45	90	3
2023103	PCC303	Database Management System	3	--	--	45	--	--	45	90	3
2023104	PCC304	Object Oriented Programming using Java	2	--	--	30	--	--	30	60	2
	OEC301	Open Elective	2	--	--	30	--	--	30	60	2
2023105	PCT301	Applied Statistics Lab	--	--	--	--	15	--	15	30	1
2023106	PCL302	Intelligent Problem Solving Lab	--	--	2	--	--	30	--	30	1
2023107	PCL303	Database Management System Lab	--	--	2	--	--	30	--	30	1
2023108	PCL304	Java Programming Lab	--	--	2	--	--	30	--	30	1
2023601	CEP301	Nirman Lab-III	--	--	2	--	--	30	--	30	1
2993501	HSL301	**Entrepreneurship Development** (Syllabus common to all Branches).	--	---	2*+2	--	--	60	--	60	2
2993502	VECL301	**Engineering for Sustainability** (Syllabus common to all Branches).	--	--	2*+2	--	--	60	--	60	2
	Total		**12**	**00**	**18**	**180**	**15**	**270**	**180**	**630**	**21**

* Two hours of practical class to be conducted for the full class as a demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Note:

1) **HSL301, and VECL301** will be common to all branches.

2) **PCC, and PCL** are Program Specific.

3) **Questions based on self-learning** must not be asked in the internal assessment and the end-of-semester exam.

4) *** Nirman Lab**: The Nirman Lab fosters individual interdisciplinary projects through theme-based, subject-integrated, experiential learning, assessed via standardised rubrics for innovation, application, and documentation."

Notional Learning Hours: In higher education, Notional Learning Hours (NLH) indicates the estimated time that a typical learner, at a specific academic level, is expected to invest in order to achieve the set learning outcomes of a course or module. This time includes a range of structured and independent learning activities, each contributing to the development of knowledge and competencies within a module.

Self-Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and

requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

Course Code	Course Type	Course Description	Examination scheme							
			Continuous Assessment (CA)			End Sem. Exam Marks (ESE)	End Sem. Exam Duration of Theory (Hrs)	Term Work (Tw)	Pract & Oral	Total
			IAT-I	IAT-II	Total= IAT-I+IAT-II					
2023101	PCC301	Applied Statistics & Probability for Data Science	20	20	40	60	02	--	--	100
2023102	PCC302	Algorithms for Intelligent Problem Solving	20	20	40	60	02	--	--	100
2023103	PCC303	Database Management System	20	20	40	60	02	--	--	100
2023104	PCC304	Object Oriented Programming using Java	20	20	40	60	02	--	--	100
	OEC301	Open Elective	20	20	40	60	02	--	--	100
2023105	PCT301	Applied Statistics Lab	--	--	--	--	--	25	--	25
2023106	PCL302	Intelligent Problem Solving Lab	--	--	--	--	--	25	25	50
2023107	PCL303	Database Management System Lab	--	--	--	--	--	25	25	50
2023108	PCL304	Java Programming Lab	--	--	--	--	--	25	25	50
2023601	CEP301	Nirman Lab-III	--	--	--	--	--	25	25	50
2993501	HSL301	**Entrepreneurship Development** (Syllabus common to all Branches).	--	--	--	--	--	50	--	50
2993502	VECL301	**Engineering for Sustainability** (Syllabus common to all Branches).	--	--	--	--	--	50	--	50
	Total		**100**	**100**	**200**	**300**	**10**	**225**	**100**	**825**

Curriculum Structure and Examination Scheme
for
Second Year Bachelor of Engineering of Computer Engineering
Terna Engineering College (With Effect from 2026-2027)

Semester-IV

Course Code	Course Type	Course Description	Teaching & Learning Scheme								
			Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per Semester)					Total Credit (C) Total Hours (Notional Learning Hours)/30
			L	T	P	L	T	P	SL	Notional Learning Hour	
2024101	PCC401	Theory of Computation	2	–	–	30	--	--	30	60	2
2024102	PCC402	Operating System	3	--	–	45	--	--	45	90	3
2024103	PCC403	Computer Organization &Architecture	3	--	--	45	--	--	45	90	3
	MDMC401	Multidisciplinary minor	3	--	--	45	--	--	45	90	3
	OEC401	Open Elective	2	--	--	30	--	--	30	60	2
2024104	PCT401	Theory of Computation Tutorial	--	1	--	--	15	--	15	30	1
2024105	PCL402	Operating System Lab	--	--	2	--	--	30	--	30	1
2024106	PCL403	Nirman Lab-IV	--	--	2	--	--	30	--	30	1
	MDML401	Multidisciplinary minor Lab	--	--	2	--	--	30	--	30	1
2024401	VSEC401	Linux Operating System	--	--	2*+2	--	--	60	--	60	2
2994501	EEM402	**Project Management** (Syllabus common to all Branches).	--	---	2*+2	--	--	60	--	60	2
2994502	VEC403	**Design Thinking** (Syllabus common to all Branch).	--	--	2*+2	--	--	60	--	60	2
	Total		**13**	**01**	**18**	**195**	**15**	**270**	**210**	**690**	**23**

* Two hours of practical class to be conducted for the full class as a demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Note:

1) **EEM402, and VEC403** will be common to all branches.

2) **PCC, and PCL** are Program Specific.

3**) MDMC, MDML, OEC** – will be as per students choice.

3) **Questions based on self-learning** must not be asked in the internal assessment and the end-of-semester exam.

4) *** Nirman Lab**: The Nirman Lab fosters individual interdisciplinary projects through theme-based, subject-integrated, experiential learning, assessed via standardised rubrics for innovation, application, and documentation."

Notional Learning Hours: In higher education, Notional Learning Hours (NLH) indicates the estimated time that a typical learner, at a specific academic level, is expected to invest in order to achieve the set learning outcomes of a course or module. This time includes a range of structured and independent learning activities, each contributing to the development of knowledge and competencies within a module.

Self-Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

Course Code	Course Type	Course Description	Examination scheme							
			Continuous Assessment (CA)			End Sem. Exam Marks (ESE)	End Sem. Exam Duration of Theory (Hrs)	Term Work (Tw)	Pract & Oral	Total
			IAT-I	IAT-II	Total=IAT-I+IAT-II					
2024101	PCC401	Theory of Computation	20	20	40	60	02	--	--	100
2024102	PCC402	Operating System	20	20	40	60	02	--	--	100
2024103	PCC403	Computer Organization &Architecture	20	20	40	60	02	--	--	100
	MDMC401	Multidisciplinary minor	20	20	40	60	02	--	--	100
	OEC401	Open Elective	20	20	40	60	02	--	--	100
2024104	PCT401	Theory of Computation	--	--	--	--	--	25	--	25
2024105	PCL402	Operating System Lab	--	--	--	--	--	25	25	50
2024106	PCL403	Nirman Lab-IV	--	--	--	--	--	25	25	50
	MDML401	Multidisciplinary minor Lab	--	--	--	--	--	25	25	50
2024401	VSEC401	Linux Operating System	--	--	--	--	--	25	25	50
2994501	EEM402	**Project Management** (Syllabus common to all Branches).	--	--	--	--	--	50	--	50
2994502	VEC403	**Design Thinking** (Syllabus common to all Branch).	--	--	--	--	--	50	--	50
	Total		**100**	**100**	**200**	**300**	**10**	**225**	**100**	**825**

S.E. Semester – III

Bachelor of Engineering Computer Engineering

Syllabus

Theme
Arogya Pranali: Intelligent Health Systems for Bharat

Focus: Ayushman Bharat | Women & Child Development

Aligned with UNSDG: SDG 3 ·

Keywords: Ayushman Bharat, PM-JAY, ABDM, ABHA, Child Health, Women Empowerment, Maternal Health, Malnutrition, ASHA, Anganwadi, POSHAN Abhiyan, Frontline Health Workers, WCD, Digital Health

Theme Description

Arogya Pranali: Intelligent Health Systems for Bharat is a focused theme centred on harnessing artificial intelligence and digital innovation to strengthen India's most critical health pillars: Ayushman Bharat and the holistic development of women and children. It calls upon innovators and technologists to co-create solutions deeply rooted in the lived realities of Bharat its villages, aspirational districts, and underserved communities.

India's Ayushman Bharat programme through PM-JAY's financial protection, Health and Wellness Centres, and the Ayushman Bharat Digital Mission (ABDM) provides the foundational infrastructure for universal health coverage. Yet millions of eligible beneficiaries remain unreached, health records stay fragmented, and last-mile delivery continues to fall short. Arogya Pranali invites solutions that make Ayushman Bharat's promise real extending its reach to the poorest households and integrating intelligent systems with the ABHA health ID and Health Information Exchange frameworks.

At the heart of this theme lies a firm commitment to child health and women's empowerment. India continues to grapple with child malnutrition, stunting, and maternal mortality challenges deeply linked to poverty and limited access to timely care. The theme invites AI-powered solutions for early detection of malnutrition, high-risk pregnancy surveillance, and vernacular digital health tools that empower ASHAs, ANMs, and Anganwadi workers as informed agents of community health.

Ultimately, Arogya Pranali envisions health systems that are predictive rather than reactive, equitable rather than exclusive, and community-centred rather than institution-bound treating every woman, child, and citizen of Bharat as an empowered participant in their own health journey.

Alignment with UNSDG 3: Good Health & Well-Being

SDG 3 aims to "ensure healthy lives and promote well-being for all at all ages." Arogya Pranali advances this goal across four pillars:

SDG 3 Target	Policy / Mission
3.8 – Universal Health Coverage	Ayushman Bharat (PM-JAY, HWC, ABDM)
3.1 – Reduce Maternal Mortality	PMMVY, RCH Portal, WCD Ministry
3.2 – End Preventable Child Deaths	POSHAN Abhiyan, ICDS, Anganwadi Network
3.4 – Reduce NCD Mortality	Health & Wellness Centres (HWC)
3.c – Strengthen Health Workforce	ASHA, ANM, Anganwadi Workers

3.8 Universal Health Coverage (PM-JAY & ABDM)

Millions of eligible PM-JAY beneficiaries particularly in aspirational districts remain unaware of their entitlements or unable to navigate the claims process. Intelligent beneficiary identification tools and vernacular outreach directly advance SDG 3.8's target of financial risk protection and access to quality essential health services for all. ABDM's ABHA health ID and Health Information Exchange enable longitudinal, consent-based records that ensure continuity of care across HWCs, district hospitals, and

tertiary centres.

3.1 & 3.2 Maternal and Child Mortality Reduction

AI-powered maternal risk scoring tools deployable by ASHAs integrated with the RCH portal predict obstetric complications and prioritise emergency referrals, directly contributing to SDG 3.1 (maternal mortality) and SDG 3.2 (child mortality) targets. Mobile-based anthropometric assessment and growth tracking tools extend malnutrition surveillance between Anganwadi visits, enabling earlier intervention for children at risk of stunting and wasting.

3.4 Non-Communicable Disease Prevention (HWCs)

Health and Wellness Centres are the frontline of comprehensive primary care, yet frequently lack decision-support tools for early NCD detection, maternal monitoring, and child health screening. AI-assisted triage and diagnostic aids enhance the clinical capacity of HWC workers, directly addressing SDG 3.4's target of reducing premature mortality from non-communicable diseases.

3.c Strengthening the Frontline Health Workforce

ASHAs, ANMs, and Anganwadi workers are the most critical link in India's health delivery chain. AI-assisted decision-support tools amplify their clinical capacity enabling real-time risk identification, smarter referrals, and improved follow-up creating both health impact and professional empowerment for this essential workforce.

Problem Statement

1. Millions of children suffer from malnutrition and stunting that remain undetected between Anganwadi visits due to the absence of continuous, community-level monitoring tools.

2. Early warning signs of obstetric complications are frequently missed in high-burden states due to under-equipped frontline workers and the absence of structured, real-time referral pathways.

3. Millions of eligible PM-JAY beneficiaries in aspirational districts remain unaware of their entitlements or are unable to navigate a claims system that was not designed for low-literacy, low-connectivity populations.

4. Despite the rollout of ABHA IDs under ABDM, the health records of pregnant women, lactating mothers, and children under five remain fragmented across paper registers and disconnected systems, making continuity of care nearly impossible.

5. Adolescent girls in aspirational districts face compounding burdens of anaemia, malnutrition, early marriage, and mental health distress that are systematically invisible to the health system due to the absence of targeted screening and data infrastructure.

6. ASHAs and Anganwadi workers are expected to manage growing responsibilities maternal surveillance, child nutrition, immunization tracking, NCD screening with paper-based tools and no real-time decision support, limiting their effectiveness in the communities they serve.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023101	Applied Statistics and Probability for Data Science	02	--	--	30	--	--	30	60	02

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023101	Applied Statistics and Probability for Data Science	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The course Applied Statistical Learning & Artificial Intelligence is designed in alignment with NEP 2020 with a strong focus on skill-based, multidisciplinary, experiential, and industry-oriented learning. The earlier conventional statistics syllabus was predominantly theoretical and lacked integration with modern computational technologies, intelligent analytics, and Artificial Intelligence-based applications. This revised curriculum integrates statistical foundations, probabilistic reasoning, predictive analytics, and intelligent data interpretation using R Programming. Aligned with the Semester III theme of intelligent computing and data-driven technologies, the course establishes a strong foundation for advanced domains such as Artificial Intelligence, Machine Learning, Deep Learning, Data Visualization, Data Mining, and Intelligent Decision Systems. The curriculum supports the vision of NITI Aayog's National Strategy for Artificial Intelligence, promoting inclusive and responsible AI adoption across sectors such as healthcare, agriculture, education, smart mobility, and governance in India. The course aligns with UNSDG 3, 4, 8, 9, 11, and 13 by encouraging students to develop data-driven and AI-enabled solutions for sustainable societal and industrial development. Through statistical learning, predictive analytics, and intelligent analytics using R Programming, students gain technological and analytical competencies that contribute toward sustainable innovation, digital transformation, and responsible intelligent systems for societal advancement.

Course Objectives:

Sr. No.	Course Objectives
1	Understand statistical foundations required for AI and Machine Learning.
2	Apply probability concepts for uncertainty handling and intelligent decision-making.
3	Analyze real-world datasets using statistical techniques.
4	Build foundational understanding for predictive analytics and machine learning.
5	Use R Programming and open-source tools for statistical computing and visualization.
6	Interpret statistical results for solving Indian societal and engineering problems.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Apply descriptive statistical methods to analyze real-world datasets	L1
2	Use probability and probabilistic reasoning in intelligent systems	L3
3	Analyze random variables and probability distributions for data-driven applications.	L4
4	Perform statistical inference and hypothesis testing for decision-making. Analyze	L4
5	Apply correlation and regression techniques for predictive analysis. Apply	L3
6	Develop data-driven insights and visualization-based interpretations for real-world applications	L6

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Mathematics Fundamentals, Computational Thinking , Basics of Programming, Data Interpretation Skills		-
1	**Statistical Foundations for Data Science**	1.1 Introduction to Applied Statistics, Role of Statistics in AI & ML, 1.2 Types of Data, Measurement Scales, Data Collection Methods, 1.3 Sampling Techniques, Data Cleaning, Measures of Central Tendency, Measures of Dispersion, Measures of Shape, Introduction to Real Indian Datasets **Self-Learning:** Data Ethics, Open Government Data Online References: https://onlinecourses.nptel.ac.in/noc21_ma01/preview	05	CO1
2	**Probability & Probabilistic Thinking**	2.1 Fundamentals of Probability, Random Experiments, Conditional Probability 2.2 Bayes' Theorem, Independent and Dependent Events, 2.3 Probabilistic Decision Making, Uncertainty in AI Systems **Self-Learning:** Bayesian Thinking in AI, Probability in Medical Diagnosis, Recommendation Systems, Spam Filtering Online References: NPTEL – Probability I with Examples using R Probability Theory for Data Science https://onlinecourses.nptel.ac.in/noc25_ma48/preview	05	CO2
3	**Random Variables & Probability Distributions**	3.1 Discrete and Continuous Random Variables, PMF, PDF, CDF, Expectation, Variance, 3.2 Binomial Distribution, Poisson Distribution, Normal Distribution, Introduction to Central Limit Theorem **Self-Learning:**	04	CO3

		Normal Distribution in ML, Telecom Data Analytics, Traffic Prediction, AI Uncertainty Modeling Online References: https://onlinecourses.nptel.ac.in/noc25_ma20/preview https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_ma20		
4	**Statistical Inference & Hypothesis Testing**	4.1 Sampling and Sampling Distribution, Confidence Intervals, Statistical Inference, 4.2 Hypothesis Testing, Null and Alternative Hypothesis, 4.3 Type I & II Errors, t-Test, Chi-Square Test, p-value Interpretation **Self-Learning:** A/B Testing, Survey Analytics, Industrial Quality Analysis, Statistical Decision Making Online References: https://onlinecourses.nptel.ac.in/noc24_ma43/preview https://onlinecourses.nptel.ac.in/e-learning/preview/noc24_ma43	06	CO4
5	**Correlation, Regression & Predictive Analytics**	5.1 Covariance, Correlation, Karl Pearson Correlation Coefficient, Scatter Plot Analysis, 5.2 Simple Linear Regression, Classification , Trend Analysis, 5.3 Basics of Predictive Modeling, Introduction to Machine Learning Perspective **Self-Learning:** Predictive Analytics for Agriculture, AI Forecasting, Correlation in Social Media Analytics, Regression in ML Online References: https://www.statlearning.com/	05	CO5
6	**Statistical Learning & Data Visualization**	6.1 Introduction to statistical learning, Supervised and unsupervised learning concepts, 6.2 Exploratory Data Analysis (EDA), Data preprocessing and feature selection, 6.3 Histograms, Scatter plots, Box plots, Heat maps, Power BI, Python libraries overview **Self-Learning:** Dashboard Design, Data Storytelling, Open Source Analytics Ecosystem, AI Data Pipelines, Tableau Online References: https://r4ds.had.co.nz/	05	CO6
		Total Hours	**30**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Bruce, P., Bruce, A., & Gedeck, P. (2020). Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python (2nd ed.). O'Reilly Media.
2. Blitzstein, J. K., & Hwang, J. (2019). Introduction to Probability (2nd ed.). CRC Press.
3. Singh, H. (2017). Statistics for Machine Learning. Packt Publishing.
4. Wickham, H., & Grolemund, G. (2016). R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media.
5. Downey, A. B. (2014). Think Stats: Exploratory Data Analysis in Python (2nd ed.). O'Reilly Media.
6. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning: With Applications in R (2nd ed.). Springer.
7. Devore, J. L. (2015). Probability and Statistics for Engineering and the Sciences (9th ed.). Cengage Learning.
8. Matloff, N. (2011). The Art of R Programming: A Tour of Statistical Software Design. No Starch Press.

Online References:

Sr. No.	Website Name	URL
1	An Introduction to Statistical Learning	https://static1.squarespace.com/static/5ff2adbe3fe4fe33db902812/t/6009dd9fa7bc363aa822d2c7/1611259312432/ISLR+Seventh+Printing.pdf
2	Statistical Computing	https://www.r-project.org
3	IDE for R	https://posit.co/download/rstudio-desktop
4	Visualization	https://ggplot2.tidyverse.org/
5	Interactive Analysis	https://jupyter.org/
6	AI & ML Programming	https://www.python.org/

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023102	Algorithms for Intelligent Problem Solving	03	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023102	Algorithms for Intelligent Problem Solving	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

In line with National Education Policy 2020 (NEP 2020), the syllabus of "Algorithms for Intelligent Problem Solving" is also aligned with the theme "Arogya Pranali Intelligent Health Systems for Bharat" and relevant UNSDG goals through its focus on computational efficiency, optimization, intelligent decision-making, healthcare data analysis, and scalable algorithmic solutions. The course enables students to apply advanced algorithmic techniques in healthcare applications such as medical image processing, ambulance routing, patient scheduling, hospital resource allocation, and medical text mining, thereby contributing toward intelligent, sustainable, and technology-driven healthcare systems.

Course Objectives:

Sr. No.	Course Objectives
1	To understand the fundamental concepts of algorithm design, performance analysis, and asymptotic notations.
2	To develop the ability to analyse the time and space complexity of various algorithms and recurrence relations.
3	To apply divide-and-conquer techniques for solving computational problems efficiently.
4	To design and implement algorithms using greedy, dynamic programming, backtracking, and branch & bound strategies.
5	To understand and apply string matching algorithms for efficient pattern searching in real-world applications.
6	To gain knowledge of computational complexity classes such as P, NP, NP-Hard, and NP-Complete and their significance in problem solving.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Analyse the efficiency of algorithms using asymptotic notations and recurrence relation methods.	L4
2	Apply divide-and-conquer strategies to solve sorting, searching, and matrix computation problems.	L3

3	Apply greedy and dynamic programming approaches for graph and optimization problems.	L3
4	Implement backtracking and branch & bound techniques for solving combinatorial and optimization problems	L3
5	Compare and apply different string matching algorithms for efficient text and pattern processing applications.	L4
6	Classify computational problems based on complexity theory concepts such as P, NP, NP-Hard, and NP-Complete.	L2

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Data structure, C programming, Concept of graphs		-
1	**Introduction and Complexity Analysis**	1.1 Performance analysis , space and time complexity, Growth of function Big Oh ,Omega , Theta notation, Mathematical background for algorithm analysis, 1.2 Analysis of selection sort , insertion sort and bubble sort. 1.3 Recurrences: The substitution method, Recursion tree method, Master method **Self-learning Topics:** Randomized algorithms, Approximation algorithms, Complexity of healthcare datasets Online reference: https://www.youtube.com/watch?v=S47aSEqm_0I https://www.youtube.com/watch?v=9Q_cJ78SQd4&t=1s https://www.youtube.com/watch?v=2wT2xOZJFx8&t=967s	08	CO1
2	**Divide and Conquer Strategy**	2.1 General method, Merge Sort, Quick Sort, Finding Minimum and Maximum algorithm and its analysis 2.2 Binary Search Analysis, Strassen Matrix Multiplication, Analysis and complexity **Self-learning Topics:** Parallel Divide and Conquer algorithms, Medical image segmentation applications Online reference: https://www.youtube.com/watch?v=KrjWloKRE2U https://www.youtube.com/watch?v=2UP-XMgOyNw&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=13 https://www.youtube.com/watch?v=Pzs15qv-xTk&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=14 https://www.youtube.com/watch?v=RajaRfMJF4I&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=15 https://www.youtube.com/watch?v=RajaRfMJF4I&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=15	07	CO2
3	**Greedy and Dynamic Programming Approaches**	3.1 Greedy Approach : Dijkstra Algorithm, Fractional Knapsack, Job Sequencing with deadlines, Prim's Algorithm, Kruskal Algorithm 3.2 Dynamic Approach: Bellman-Ford Algorithm, Floyd Warshall Algorithm, 0/1 Knapsack, Longest Common Subsequence, Multistage graphs	09	CO3

		Self-learning Topics: Route optimization for ambulance systems, Genome sequence alignment Online reference: https://www.youtube.com/watch?v=EcT-Jt5WStw&list=PL7DC83C6B3312DF1E&index=10 https://www.youtube.com/watch?v=nieg8XQQ7i0&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=25 https://www.youtube.com/watch?v=-M6IzZMpm1Y&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=29 https://www.youtube.com/watch?v=b2-vAkasvD0&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=30 https://www.youtube.com/watch?v=YbbGemzQ5ok&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=26 https://www.youtube.com/watch?v=_4CGMB5j_CQ&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=27 https://www.youtube.com/watch?v=ml0i13Uvmuc&list=PL7DC83C6B3312DF1E&index=19		
4	**Backtracking and Branch & Bound**	4.1 Backtracking: N-Queen Problem, Sum of Subsets, Graph Coloring 4.2 Branch & Bound : Travelling Salesperson Problem, 15 Puzzle Problem **Self-learning Topics:** Patient scheduling optimization, Hospital resource allocation systems Online reference: https://www.youtube.com/watch?v=0PyutiOvIkU https://www.youtube.com/watch?v=kdBzkxdJ7bI&t=477s https://www.youtube.com/watch?v=-cLsEHP0qt0&t=16s	06	CO4
5	**String Matching**	5.1 Naïve String Matching, Rabin-Karp Algorithm, Finite Automata String Matching, Knuth-Morris-Pratt Algorithm, **Self-learning Topics:** Medical text mining and symptom extraction systems, text search in documents Online reference: https://www.youtube.com/watch?v=EEjNb9yUv1k&t=53s https://www.youtube.com/watch?v=RYUGTzI-f-A&pp=ygUZU3RyaW5nIE1hdGNoaW5nICBieSBucHRlbA%3D%3D https://www.youtube.com/watch?v=8j9wOsI4cx8 https://www.youtube.com/watch?v=NeVqZ5YFmgs	08	CO5
6	**Computational Complexity**	6.1 Complexity class: Definition of P, NP, NP-Hard, NP-Complete 6.2 Reducibility, Vertex Cover, Clique problem **Self-learning Topics:** Complexity Analysis of Healthcare Decision Support Systems, NP-Complete Problems in Healthcare Optimization, Computational Complexity in Artificial Intelligence and Machine Learning, Approximation Algorithms for Real-Time Healthcare Systems Online reference:	07	CO6

		https://www.youtube.com/watch?v=bE4d4IiOEQk https://www.youtube.com/watch?v=iBX_3wLreWc&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=55		
			Total Hours	**45**

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, "Introduction to algorithms", 2nd Edition, PHI Publication 2005.
2. Ellis Horowitz, Sartaj Sahni, S. Rajsekaran. "Fundamentals of computer algorithms", University Press.
3. Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, "Algorithms", Tata McGraw- Hill Edition.
4. S. K. Basu, "Design Methods and Analysis of Algorithm", PHI

Online References:

Sr. No.	Website Name	URL
1	Nptel Swayam	nptel.ac.in/courses/106106131
2	Nptel Swayam	https://nptel.ac.in/courses/106105085?
3	Coursera	https://www.coursera.org/specializations/data-structures-algorithms?

Assessment:

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End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023103	Database Management System	3	-	-	45	-	-	45	90	3

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023103	Database Management System	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The proposed DBMS syllabus is strategically designed to support India's digital healthcare transformation by positioning data management as a foundation for intelligent health systems. Aligned with NEP 2021, it emphasizes experiential learning, interdisciplinary problem-solving, and the integration of emerging database technologies. Core concepts such as ER modelling, normalization, transactions, concurrency control, and recovery are taught using healthcare-focused applications and case studies. The curriculum develops the skills required to design secure, scalable, and reliable health information systems that address real-world societal challenges. Advanced and self-learning modules on cloud, distributed, and NoSQL databases further strengthen alignment with the Ayushman Bharat Digital Mission (ABDM) by enabling healthcare data interoperability and large-scale digital health solutions. Overall, approximately 80% of the syllabus aligns with the "Arogya Pranali—Intelligent Health Systems for Bharat" theme and contributes to SDGs 3, 5, and 9, while the remaining content provides essential theoretical foundations for database professionals.

Course Objectives:

Sr. No.	Course Objectives
1	Analyse file-based system limitations vs. DBMS advantages.
2	Design conceptual models using ER/EER.
3	Apply relational algebra and normalization to reduce redundancy, identify keys, and decompose relations with non-additive joins.
4	Construct and execute SQL and explain query processing, optimization, indexing, joins, and cost estimation.
5	Compare ACID, concurrency control, deadlock handling, recovery, security (views, row/column-level), and healthcare compliance.
6	Differentiate RDBMS, NoSQL types, distributed databases cloud models, and MongoDB features.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Understand database system architecture with its components, data independence, and their role in healthcare applications.	L2
2	Design ER/EER models incorporating entities, relationships, constraints, specialization/generalization, and transform them into relational schemas.	L3
3	Formulate relational algebra queries and normalize database tables using functional dependencies, candidate key identification, and decomposition rules.	L3 - L4

4	Write complex SQL queries and Examine query processing and optimization techniques.	L3
5	Analyse transaction management mechanisms and database security measures for compliant data management.	L4
6	Compare and apply relational, distributed, cloud, and NoSQL database models.	L2–L3

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisite**	Computer Fundamental and Programming, Foundational Mathematics, Conceptual Understanding	0	-
1	**Introduction to Database Systems and ER modeling**	1.1 Introduction to DBMS vs file systems, 1.2 three layer abstraction architecture, data independence, 1.3 Characteristics of database systems, 1.4 Different types of Database users and their roles, 1.5 DBMS Architecture, 1.6 Role of Database in Healthcare. 1.7 ER modelling: Entity, Entity Set, Attributes and its types, relationship, 1.8 relationship set & constraints , 1.9 Keys and Data integrity constraints: domain, entity, referential with foreign key, 1.10 specialization, generalization, aggregation 1.11 constraints of specialization and generalization. **Self-learning Topics:** Role of databases in different applications. Difference between hierarchical, network and relational database models. Online reference: Preview: Data Base Management System | SWAYAM https://www.coursera.org/programs/e-band-competency-learning-program-ll4xm/learn/illinois-tech-data-modeling-and-applications?authProvider=atlnext	10	CO1,CO2
2	**Relational Model, Algebra and Normalization**	2.1 Relational Model: terminology, 2.2 Mapping ER/EER to relational schema. 2.3 Relational algebra and operations, 2.4 Relational algebra queries, 2.5 Need of normalization, Functional dependencies, 2.6 Normal Forms: Candidate key identification, 1NF, 2NF, 3NF, BCNF and 4NF, 2.7 decomposition rules. **Self-learning Topics:** Inference Rules for Functional Dependencies, Nonaddictive join test for decomposition. Online reference: https://brainkart.com/article/Further-Topics-in-Functional-Dependencies--Inference-Rules--Equivalence--and-Minimal-Cover_11503 https://turing.une.edu.au/~cosc210/lectures/lecture_16/Lecture_16.pdf	09	CO2,CO3
3	**SQL , Query Processing & Optimization**	3.1 Introduction to SQL, 3.2 DDL, DML, DCL, TCL commands, 3.3 Constraints and views, 3.4 Joins and nested queries, 3.5 Aggregate and string functions, 3.6 Stored procedures and functions, 3.7 Triggers and cursors, 3.8 PL/SQL blocks. 3.9 Overview of Query Processing, 3.10 Query Parsing and Validation,	08	CO4

		3.11 Relational Algebra Translation , 3.12 Query Optimization, 3.13 Query Execution Plans, 3.14 Indexing Strategies: Multi-column Indexes, 3.15 Join Algorithms, Sorting and Aggregation, 3.16 Cost Estimation, 3.17 Materialized Views, 3.18 Query Rewriting. **Self-learning Topics:** Adaptive Query Optimization, Machine Learning for Query Optimization, Healthcare-Specific Query Optimization. Online reference: https://www.cse.iitb.ac.in/infolab/ https://www.cse.iitb.ac.in/infolab/		
4	**Transactions, Concurrency and Security**	4.1 Transaction Concept: Transaction States, ACID properties, 4.2 Concurrency Control: Schedules, Serializability, Locking Protocols, Timestamp Based Protocol, 4.3 Deadlock Handling, 4.4 Types of Failures: Transaction Failure, System Failure, Media Failure and Application Errors, 4.5 Database Recovery Techniques, 4.6 Database Security: Authorization and Access Control, View-based access control, Column/Row-level security. **Self-learning Topics:** Data privacy and ethical healthcare data handling, Auditing & Compliance (HIPAA, GDPR for healthcare), SQL Injection prevention. Online reference: https://www.oracle.com/in/security/database-security/	07	CO5
5	**Distributed database Systems, Cloud Databases**	5.1 Limitations of traditional RDBMS, 5.2 Distributed Database System: Architecture Types, fragmentation and replication, Distributed Query Processing. 5.3 Cloud Database: concepts, Database-as-a-Service (DBaaS), Multi- cloud Database Strategies. **Self-learning Topics:** AWS HealthLake, Azure for Healthcare**,** Serverless databases (Aurora Serverless, Snowflake, BigQuery), Multi-region active-active databases (Spanner, Cosmos DB). Online reference: https://aws.amazon.com/healthlake/ **https://learn.microsoft.com/en-us/azure/healthcare-apis/** https://docs.snowflake.com/en/ https://learn.microsoft.com/en-us/azure/cosmos-db/how-to-multi-master?tabs=api-async	05	CO6
6	**NoSql Database**	6.1 Introduction to NoSQL databases. 6.2 Types of NoSQL database patterns: Document databases, Key-value stores, Column-family databases, Graph databases, 6.3 MongoDB: Documents, collections, CRUD, aggregation pipeline, indexing, replica sets, sharding, change streams. **Self-learning Topics:** Federate database architecture, Firebase, ACID vs CAP theorem comparison. MongoDB with Cloud Platforms	06	CO6

		Online reference: https://www.ibm.com/docs/en/db2/11.1.0?topic=federation-federated-systems https://firebase.google.com/docs https://dba.stackexchange.com/questions/31260/consistency-in-acid-and-cap-theorem-are-they-the-same https://www.mongodb.com/resources/basics/databases/cloud-databases/free-cloud-database		
			Total Hours	45

Note:
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References:

1. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", 7th Edition, Pearson.
2. Pramod J. Sadalage, Martin Fowler "NoSQL Distilled", Addison-Wesley.
3. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, "MongoDB: The Definitive Guide", 3rd Edition, O'Reilly.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://onlinecourses.nptel.ac.in/e-learning/preview/noc24_cs75
2	Coursera	https://www.coursera.org/programs/e-band-competency-learning-program-ll4xm/learn/illinois-tech-data-modeling-and-applications?authProvider=atlnext
3	IITB	https://www.cse.iitb.ac.in/infolab/
4	MongoDB	https://www.mongodb.com/resources/basics/databases/cloud-databases/free-cloud-database

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End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	2
2023104	Object Oriented Programming using Java	02	--	--	30	--	--	30	60	

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023104	Object Oriented Programming using Java	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The healthcare ecosystem of Bharat is rapidly evolving through digital initiatives such as the Ayushman Bharat Digital Mission, telemedicine platforms, AI-assisted diagnostics, wearable health monitoring systems, and smart healthcare services. Developing such intelligent and scalable healthcare solutions requires strong foundations in Object-Oriented Programming (OOP), software engineering principles, and robust application development practices. Aligned with the vision of NEP 2020, this course promotes experiential learning, problem-solving, critical thinking, innovation, multidisciplinary application development, and industry-relevant skill acquisition. The curriculum enables students to apply Java programming concepts in designing modular, secure, and reusable software components for healthcare applications such as Electronic Health Records (EHR), hospital management systems, remote patient monitoring, pharmacy management, and healthcare analytics. The course contributes to the achievement of UNSDG goals 3, 9 and 10. Through the theme of intelligent healthcare systems for Bharat, students develop competencies required to build sustainable and socially impactful digital solutions for the healthcare sector.

Course Objectives:

Sr. No.	Course Objectives
1	To transition students from procedural programming logic to object-oriented thinking and design patterns.
2	To explain the architecture of the Java Virtual Machine (JVM), including memory management, bytecode, and platform independence.
3	To provide in-depth knowledge of the four pillars of OOP: Encapsulation, Inheritance, Polymorphism, and Abstraction.
4	To enable students to build reliable applications using exception handling and multithreading mechanisms.
5	To teach the use of packages, interfaces, and abstract classes for creating modular and loosely-coupled code.
6	To introduce fundamental data structures (Collections) and I/O streams for efficient data handling and persistence.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the architectural differences between procedural and object-oriented paradigms, including the role of JVM, JRE, and JDK in platform independence.	L2
2	Model real-world entities as Java classes by applying encapsulation, constructors, access modifiers, and static members.	L3
3	Implement code reusability and dynamic behavior using inheritance hierarchies, method overriding, and runtime polymorphism.	L3
4	Design decoupled and scalable system architectures by utilizing abstract classes, interfaces, and packages.	L6
5	Construct fault-tolerant applications by implementing structured exception handling (try-catch-throw) and user-defined exceptions.	L6
6	Utilize wrapper classes, Collections framework (ArrayList, HashMap), and basic file I/O streams for efficient data storage and manipulation.	L3

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Foundations of C Programming (Variables, Control Structures - if/else/switch, Loops - for/while/do-while, Functions, Arrays)	02	-
1	**Introduction to OOP & Java Basics**	1.1. Evolution of Programming: Procedural vs. Object-Oriented Paradigms. 1.2. OOP Concepts: Objects, Classes, Encapsulation, Abstraction, Inheritance, Polymorphism, Message Passing. 1.3. Java Architecture: JVM (Memory Management - Stack/Heap), JRE, JDK, Bytecode, WORA (Write Once Run Anywhere). 1.4. Structure of a Java Program (Class & Main method). 1.5. Data Types, Variables, Literals, and Type Conversion. 1.6. Operators (Arithmetic, Relational, Logical, Bitwise, Unsigned Right Shift >>>). 1.7. Control Statements (if-else, switch) & Loops (for, while, do-while, enhanced for-each). 1.8. Arrays: Declaration, Instantiation, Initialization, Multidimensional Arrays. **Self-learning Topics:** Healthcare Context Activities Patient queue management,Vital sign monitoring arrays,Disease tracking dashboards,Healthcare Information Systems Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://openmrs.atlassian.net/wiki/spaces/docs/overview https://www.openmhealth.org	04	CO1
2	**Classes, Objects, and Encapsulation**	2.1. Defining Classes and Creating Objects. 2.2. Data Members (Instance/Static) and Member Methods. 2.3. Access Modifiers: private, default, protected, public. 2.4. Constructors: Default, Parameterized, Copy Constructors; Constructor Overloading & Chaining (this()). 2.5. this Keyword (referencing current object).	10	CO2

		2.6. static Keyword: Static variables, Static methods, Static blocks. 2.7. Garbage Collection & finalize() method. 2.8. Method Overloading (Compile-time Polymorphism). 2.9. Command Line Arguments. **Self-learning Topics:** Healthcare Context Activities Electronic Health Record (EHR) object design, Appointment scheduling system Smart prescription generator, Object-Oriented Analysis in Healthcare Online reference**:** https://www.javatpoint.info https://www.w3schools.com/java/ https://specifications.openehr.org/releases/BASE/development/architecture_overview.html https://github.com/openemr/openemr		
3	**Inheritance and Polymorphism**	3.1. Inheritance: extends keyword. 3.2. Types of Inheritance (Single, Multilevel, Hierarchical),Diamond Problem . 3.3. Method Overriding (Runtime Polymorphism). 3.4. super Keyword (to access parent class members). 3.5. Dynamic Method Dispatch. 3.6. final Keyword (variable, method, class). 3.7. Object Class: toString(), equals(), hashCode(). **Self-learning Topics:** Healthcare Context Activities, ICU patient specialization, Generalized hospital staff hierarchy Disease classification systems, Healthcare Workflow Systems, Medical Device Software Reusability. Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://openmrs.atlassian.net/wiki/spaces/docs/overview https://www.gnuhealth.org	4	CO3
4	**Abstraction, Packages, and Wrapper Classes**	4.1. Abstract Classes: Abstract methods vs Concrete methods. 4.2. Interfaces: Defining contracts using interface and implements. 4.3. Multiple Inheritance via Interfaces. 4.4. Default and Static methods in Interfaces (Java 8+). 4.5. Packages: package statement, import statement, Directory structure, Creating user-defined packages. 4.6. Wrapper Classes: Integer, Double, Character, Boolean; Autoboxing and Unboxing. **Self-learning Topics:** Interoperable telemedicine APIs Modular laboratory systems Smart wearable device interfaces Interoperability Standards in Healthcare Modular Healthcare Architecture Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://build.fhir.org/ https://developer.android.com/health-and-fitness/guides/health-connect	04	CO4

5	**Exception Handling**	5.1. Exception Hierarchy: Throwable, Error, Exception. 5.2. Checked vs Unchecked (Runtime) Exceptions. 5.3. Exception Handling Keywords: try, catch, finally, throw, throws. 5.4. Multiple Catch Blocks and Multi-catch (Java 7+). 5.5. Nested try blocks. 5.6. Creating Custom/User-defined Exceptions. **Self-learning Topics:** Healthcare Context Activities Invalid patient record handling Drug dosage validation exceptions Emergency alert handling Software Reliability in Healthcare Medical Software Risk Management Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://pmc.ncbi.nlm.nih.gov/articles/PMC8902762/ https://journals.sagepub.com/doi/10.1177/14604582241252763	03	CO5
6	**I/O Streams and Introduction to Collections**	6.1. Java I/O Streams: Byte Streams (FileInputStream, FileOutputStream) vs Character Streams (FileReader, FileWriter). 6.2. Buffered Streams (BufferedReader, BufferedWriter). 6.3. Introduction to Collections Framework: Need for dynamic data storage. 6.4. ArrayList (Dynamic arrays) vs Standard Arrays. 6.5. HashMap (Key-Value pairs) and HashSet. 6.6. Iterating using Iterator interface and Enhanced For-Loop. **Self-learning Topics:** Healthcare Context Activities Patient record persistence Hospital inventory tracking Disease outbreak data analysis Health Data Management Systems Healthcare Big Data Basics Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://github.com/openemr/openemr https://build.fhir.org/resourcelist.html	05	CO6
		Total Hours	**30**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. H. Schildt and D. Coward, “Java: The Complete Reference”, 13th ed., McGraw Hill Education, 2025.
2. K. Sierra, B. Bates, and T. Gee, “Head First Java”, 3rd ed., O'Reilly Media, 2022.
3. C. S. Horstmann, “Core Java Volume I: Fundamentals”, 13th ed., Pearson (Oracle Press), 2024.
4. E. Balagurusamy, “Object Oriented Programming with Java”, 6th ed., McGraw Hill Education India, 2019.

5. P. J. Deitel and H. M. Deitel, “Java: How to Program”, 11th ed., Pearson, 2018.
6. J. Kvedar, “Digital Health: Connecting and Empowering Patients and Clinicians”, 1st ed., Wiley, 2021.

Online References:

Sr. No.	Website Name	URL
1	Oracle	https://docs.oracle.com/javase/tutorial/ https://openjdk.org/ https://www.geeksforgeeks.org/java/
2	Govt	https://abdm.gov.in https://openmrs.org https://www.hl7.org
3	W3Schools (Java)	https://www.w3schools.com/java/
4	Javatpoint	https://www.javatpoint.info

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023105	Applied Statistical Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023105	Applied Statistical Lab	---	---	---	---	---	25		25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The Applied Statistical Learning & AI Laboratory course is designed to equip learners with practical skills in statistical analysis, data interpretation, and AI-driven decision-making using modern computational tools such as R and RStudio. The course supports the vision of NEP 2020 by promoting multidisciplinary learning, critical thinking, problem-solving abilities, data literacy, and hands-on competency development required for emerging digital professions.

Through the analysis of healthcare, social, and public datasets, students gain exposure to the principles of Arogya Pranali: A Multidisciplinary Ecosystem, where data-driven intelligence supports preventive healthcare, disease monitoring, healthcare analytics, and evidence-based policy decisions. The course also introduces learners to statistical methods that form the foundation of modern AI applications in healthcare, smart cities, and sustainable development. Furthermore, the laboratory contributes to the achievement of United Nations Sustainable Development Goals (UNSDGs), particularly SDG 3,4,9 and 11 through analytical solutions for smart city applications. By combining statistical learning with AI-oriented data analysis, the course prepares students to become responsible professionals capable of addressing societal challenges through ethical, sustainable, and intelligent use of data.

Lab Objectives:

Sr. No.	Lab Objectives
1	Understand statistical analysis using R Programming.
2	Apply probability and distribution concepts using computational methods.
3	Perform statistical inference and hypothesis testing using datasets. Develop analytical thinking for AI and data-driven applications.
4	Analyse relationships among variables using correlation and regression.
5	Visualize and interpret real-world datasets using statistical tools.
6	Develop analytical thinking for AI and data-driven applications.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Implement statistical operations using R.	L1
2	Apply probability concepts computationally.	L2
3	Analyze datasets using statistical distributions.	L3
4	Perform hypothesis testing and inference using R.	L4

5	Develop predictive models using various techniques.	L5
6	Visualize and interpret analytical results effectively.	L6

Hardware & Software Requirements:

Hardware Requirement: Desktop/Laptop Computer,Intel Core i3 Processor or higher (AMD equivalent),Minimum 4 GB RAM (8 GB recommended)	**Software Requirement:**Operating System: Windows 10/11, Linux, or macOS, R Programming Language (Latest Stable Version),RStudio IDE Required R Packages: ggplot2, dplyr, tidyverse, readr, stats, caret (or equivalent)

Detailed Syllabus:

Sr. No.	Module	Detailed Content	Hours	LO Mapping
	0	Prerequisite	2	
I	**Statistical Foundations for Data Scienc**	1.1 Introduction to R Programming, RStudio environment, data types, vectors, matrices, lists, data frames, importing datasets. **Self-learning Topics:** Installation of R Packages and Libraries, R Markdown Basics, Introduction to R Help System and Documentation Working with CSV and Excel Files Online References: https://www.coursera.org/specializations/r-programming-data-science https://www.coursera.org/specializations/data-science-r-programming-tidyverse https://www.udemy.com/course/r-programming	2	LO1
II	**Probability & Probabilistic Thinking**	2.1 Descriptive statistics, measures of central tendency and dispersion, graphical representation of data. **Self-learning Topics:** Data Cleaning Techniques, Missing Value Handling Advanced Graphical Representations, Statistical Visualization for Business and Healthcare Data Online references: https://www.coursera.org/specializations/data-science-r https://www.udemy.com/course/data-science-and-machine-learning-with-r-from-a-z https://www.udemy.com/course/data-science-and-machine-learning-with-r-from-a-z/	2	LO1, LO2
III	**Random Variables & Probability Distributions**	3.1 Probability concepts, conditional probability, Bayes theorem, random variables and probability distributions. **Self-learning Topics:** Probability Simulation using R, Monte Carlo Simulation Basics, Probability Applications in AI Systems, Random Number Generation Techniques Online References: https://www.coursera.org/specializations/statistics-with-r https://www.coursera.org/specializations/data-science-r https://www.udemy.com/course/r-programmin	3	LO2, LO3

IV	**Random Variables & Probability Distributions**	4.1 Sampling, confidence intervals, hypothesis testing, t-test, chi-square test. **Self-learning Topics:** Real-World Survey Data Analysis, A/B Testing Concepts, Industrial Quality Analysis using Statistical Methods, Hypothesis Testing in Healthcare Analytics Online References: https://www.coursera.org/specializations/statistics-with-r https://www.coursera.org/specializations/data-science-r https://www.coursera.org/specializations/data-science-r	3	LO4
V	**Correlation, Regression & Predictive Analytics**	5.1 Correlation, covariance, regression analysis, predictive analytics basics. **Self-learning Topics:** Regression Applications in Predictive Analytics, Trend Analysis for Smart Cities and Agriculture, Introduction to Classification Concepts, Correlation Analysis in Social Media Data Online References: https://www.coursera.org/specializations/r-programming-data-science https://www.coursera.org/specializations/data-science-r https://www.udemy.com/course/data-science-and-machine-learning-with-r-from-a-z/	3	LO5
VI	**Statistical Learning & Data Visualization**	6.1 Data visualization, exploratory data analysis, statistical learning concepts using Indian datasets. **Self-learning Topics:** Dashboard Design Basics, Exploratory Data Analysis on Indian Open Datasets, Data Storytelling Techniques, Introduction to AI-Oriented Data Analytics Workflows Online References: https://www.coursera.org/specializations/data-science-r-programming-tidyverse https://www.udemy.com/course/r-programming/	2	LO6

References:

1. Wickham, H., & Grolemund, G. (2023). “R for Data Science” (2nd Edition). O'Reilly Media, Inc.
2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021).” An Introduction to Statistical Learning: With Applications in R” (2nd Edition). Springer.
3. Grolemund, G. (2014). “Hands-On Programming with R: Write Your Own Functions and Simulations." O'Reilly Media, Inc

Online References:

Sr. No.	Website Name	URL
1	R Project for Statistical Computing	R: The R Project for Statistical Computing
2	R Studio	RStudio IDE | The Premier Free IDE for R
3	GgPlot	ggplot2 package - RDocumentation

Suggested List of Experiments:

Sr. No.	Experiment Name	Hours
1	Installation and setup of R and RStudio.	2
2	Perform operations on vectors, matrices, and arrays in R.	2
3	Create and manipulate data frames using R.	2
4	Import and preprocess Indian open datasets.	2
5	Compute measures of central tendency and dispersion.	2
6	Visualize datasets using bar chart, histogram, pie chart, and boxplot.	2
7	Implement probability experiments using R.	2
8	Simulation of conditional probability and Bayes theorem.	2
9	Analyze discrete and continuous probability distributions.	2
10	Perform normal distribution and random variable analysis.	2
11	Perform t-test using datasets	2
12	Perform chi-square test using datasets.	2
13	Implement correlation and covariance analysis.	2
14	Develop simple linear regression models.	2
15	Perform exploratory data analysis and visualization using ggplot2.	2

Suggested List of Assignment:

Sr No	List of Assignments / Tutorials	Hours
01	Analyse a real-world Indian dataset and compute descriptive statistics using R	2
02	Implement probability and conditional probability problems using R Programming	2
03	Study and visualize different probability distributions using R	2
04	Perform hypothesis testing on survey or healthcare datasets	2
05	Develop a simple regression model for prediction analysis	2
06	Create graphical visualizations using ggplot2 for smart city or agriculture datasets	2
07	Comparative analysis of datasets using correlation and trend analysis	2
08	Mini analytical case study using exploratory data analysis and statistical interpretation	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023106	Intelligent Problem solving Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023106	Intelligent Problem solving Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

In line with National Education Policy 2020 (NEP 2020), the syllabus of "Algorithms for Intelligent Problem Solving" is also aligned with the theme "Arogya Pranali—Intelligent Health Systems for Bharat" and relevant UNSDG goals through its focus on computational efficiency, optimization, intelligent decision-making, healthcare data analysis, and scalable algorithmic solutions. The course enables students to apply advanced algorithmic techniques in healthcare applications such as medical image processing, ambulance routing, patient scheduling, hospital resource allocation, and medical text mining, thereby contributing toward intelligent, sustainable, and technology-driven healthcare systems.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand the fundamental concepts of algorithm design and complexity analysis.
2	To implement sorting and searching algorithms using different design strategies.
3	To apply Divide and Conquer, Greedy, and Dynamic Programming techniques for solving computational problems.
4	To develop solutions using Backtracking and Branch & Bound methods for optimization problems.
5	To implement pattern matching algorithms for efficient text processing and data searching.
6	To analyze and apply algorithmic approaches in healthcare and intelligent system applications.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain algorithmic concepts and analyze time and space complexity of algorithms.	L4
2	Implement sorting and searching algorithms using Divide and Conquer techniques.	L3
3	Solve optimization problems using Greedy approaches.	L3
4	Solve optimization problems using Dynamic Programming approaches.	L3
5	Demonstrate Backtracking and Branch & Bound techniques for computational problems.	L3

6	Analyse and compare the performance of different string matching algorithms	L4

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Computer System with Intel i5 or higher processor Minimum 8 GB RAM and 100 GB Storage Keyboard, Mouse, and Internet Connectivity	Windows/Linux Operating System GCC Compiler / Code::Blocks / VS Code Web Browser for Online Learning Resources

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	**Prerequisite**	Data structure, C programming, Concept of graphs	02	
1	**Introduction and Complexity Analysis**	1.1 Growth of function Big Oh ,Omega , Theta notation 1.2 Analysis of selection sort , insertion sort and bubble sort. **Self-learning Topics:** Randomized algorithms, Approximation algorithms, Complexity of healthcare datasets Online reference: https://www.youtube.com/watch?v=1nAgMQSc-4Q&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=11 https://www.youtube.com/watch?v=_EV6xAXf5FA&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=12 https://www.youtube.com/watch?v=CATsie6bnfI&pp=ygUUYnViYmxlIHNvcnQgYnkgbnB0ZWw%3D	04	LO1
2	**Divide and Conquer Strategy**	2.1 General method, Merge Sort, Quick Sort, Finding Minimum and Maximum algorithm and its analysis, 2.2 Binary Search Analysis, **Self-learning Topics:** Parallel Divide and Conquer algorithms, Medical image segmentation applications Online reference: https://www.youtube.com/watch?v=2UP-XMgOyNw&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=13 https://www.youtube.com/watch?v=OOZxxIP5890&list=PLw8EtCUY2HiDppGSjj4IIOF8bcLhH-d3w&index=3] https://www.youtube.com/watch?v=esvCwpyPh90&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=38	06	LO2
3	**Greedy Programming Approach**	3.1Greedy Approach : Dijkstra Algorithm, Fractional Knapsack, Job Sequencing with deadlines, Prim's Algorithm, Kruskal Algorithm, **Self-learning Topics:** Route optimization for ambulance systems Online reference: https://www.youtube.com/watch?v=nieg8XQQ7i0&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=25	06	LO3

		https://www.youtube.com/watch?v=-M6IzZMpm1Y&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=29 https://www.youtube.com/watch?v=b2-vAkasvD0&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=30		
4	**Dynamic Programming Approach**	4.1 Dynamic Approach: Bellman-Ford Algorithm, Floyd Warshall Algorithm, 0/1 Knapsack, Longest Common Subsequence, Multistage graphs **Self-Learning Topics:** Genome sequence alignment Online reference: https://www.youtube.com/watch?v=_4CGMB5j_CQ&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=27 https://www.youtube.com/watch?v=YbbGemzQ5ok&list=PLyqSpQzTE6M9DKhN7z2fOpKTJWu-639_P&index=26 https://www.youtube.com/watch?v=ml0i13Uvmuc&list=PL7DC83C6B3312DF1E&index=19	06	LO4
5	**Backtracking and Branch & Bound**	5.1 Backtracking: N-Queen Problem, Sum of Subsets, Graph Coloring, 5.2 Branch & Bound : Travelling Salesperson Problem, 15 Puzzle Problem **Self-learning Topics:** Patient scheduling optimization, Hospital resource allocation systems **Online reference:** https://www.youtube.com/watch?v=0PyutiOvIkU https://www.youtube.com/watch?v=yDjadhjQ3eE	04	LO5
6	**String Matching**	6.1 Naïve String Matching, Rabin-Karp Algorithm, 6.2 Finite Automata String Matching, Knuth-Morris-Pratt Algorithm **Self-learning Topics:** Medical text mining and symptom extraction systems Online reference: https://www.youtube.com/watch?v=8j9wOsI4cx8 https://www.youtube.com/watch?v=NeVqZ5YFmgs	04	LO6
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, “Introduction to algorithms”, 2nd Edition, PHI Publication 2005.
2. Ellis Horowitz, Sartaj Sahni, S. Rajsekaran. “Fundamentals of computer algorithms” University Press.
3. Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, “Algorithms”, Tata McGraw- Hill Edition.
4. S. K. Basu, “Design Methods and Analysis of Algorithm”, PHI

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://nptel.ac.in/courses/106/106/106106131/
2	SWAYAM/NPTEL	https://swayam.gov.in/nd1_noc19_cs47/preview
3	Coursera	https://www.coursera.org/specializations/algorithms
4	Mooc	https://www.mooc-list.com/tags/algorithms

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	Implementation and analysis of Selection Sort, Insertion Sort	2
02	Implementation and analysis of Binary Search and Merge Sort using Divide and Conquer approach	2
03	Implementation and analysis of Quick Sort and Finding Minimum & Maximum using Divide and Conquer approach	2
04	Implementation and analysis of Fractional Knapsack using Greedy approach	2
05	Implementation and analysis of Job Sequencing with Deadlines.	2
06	Implementation and analysis of Minimum Cost Spanning Tree (Prim's/Kruskal's Algorithm)	2
07	Implementation and analysis of Bellman-Ford and Floyd-Warshall Algorithms using Dynamic Programming	2
08	Implementation and analysis of 0/1 Knapsack using Dynamic Programming	2
09	Implementation and analysis of Longest Common Subsequence (LCS) using Dynamic Programming	2
10	Implementation and analysis of Multistage Graph Problem	2
11	Implementation and analysis of Graph Coloring Problem	2
12	Implementation and analysis of Sum of Subsets using Backtracking techniques	2
13	Implementation and analysis of 15 Puzzle Problem using Branch & Bound techniques	2
14	Implementation and analysis of Naïve String Matching, Rabin-Karp Algorithms	2
15	Implementation and analysis of Knuth-Morris-Pratt (KMP) Algorithms	2

Suggested List of Assignment:

Sr No	List of Assignments / Tutorials	Hours
01	Study for Growth of function and Reccurance relation	2
02	Study of Finding Minimum and Maximum algorithm and its analysis and Strassen Matrix Multiplication	2
03	Compare Greedy and Dynamic Programming approaches with suitable examples and discuss their real-world applications	2
04	Explain the Graph Coloring problem and demonstrate the coloring of a given graph using the Backtracking technique.	2
05	Compare Naïve String Matching and Rabin-Karp Algorithms with suitable examples.	2
06	Differentiate between P, NP, NP-Hard, and NP-Complete problems with suitable examples.	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023107	Database Management System Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023107	Database Management System Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The DBMS Laboratory curriculum is fully aligned with NEP 2020's focus on experiential, multidisciplinary learning by centering all experiments on the theme "Arogya Pranali: Intelligent Health Systems for Bharat," where students design, query, and secure a Healthcare Management System (Patients, Doctors, Appointments) using SQL and MongoDB. Over 85–90% of the syllabus directly maps to this theme, covering DDL with integrity constraints, DML with audit logging, role-based views and access control (GRANT/REVOKE), complex joins, subqueries, PL/SQL procedures for appointment booking, triggers for patient deletion auditing, cursors for follow-up reminders, and MongoDB aggregations for diagnosis analytics. This healthcare-centric approach directly contributes to UNSDG goal 3, 9 and 16 thereby preparing students to build secure, intelligent, and inclusive digital health platforms for Bharat.

Lab Objectives:

Sr. No.	Lab Objectives
1	To enable students to design and implement relational database schemas using DDL statements with appropriate integrity constraints.
2	To develop proficiency in performing data manipulation operations and implementing row-level auditing mechanisms for data compliance.
3	To equip students with skills to formulate complex SQL queries.
4	To train students in developing stored procedures, functions, and triggers to automate business logic, enforce data integrity, and implement event-driven auditing in healthcare scenarios.
5	To introduce document-based NoSQL data modelling and enable students to perform CRUD operations and build aggregation pipelines for real-time analytics in MongoDB.
6	To familiarize students with database performance optimization techniques and conceptual understanding of replication and sharding for high availability and horizontal scaling.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Design normalized relational database schemas with a integrity constraints for a given real-world application	L6
2	Apply DML operations, implement row-level auditing using log tables with timestamps and user information, and enforce access control through views.	L3

3	Formulate and execute complex SQL queries to retrieve and manipulate data effectively.	L3
4	Develop and execute stored procedures, user-defined functions, and triggers with proper error handling.	L6
5	Construct document-based schemas in MongoDB.	L6
6	Create and analyse single and compound indexes using explain () to optimize query performance, and explain the concepts of replication and sharding.	L4

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
8GB RAM minimum, 500GB (SSD)disk space, 64-bit processor	Oracle Database 19c/21c (XE) or MySQL 8.0 or PostgreSQL 14+ MongoDB Community Edition 6.0+ or MongoDB Atlas (Cloud) /SQL Developer (Oracle), MongoDB Shell (mongosh), Compass**.**

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	**Prerequisites**	Basic Computer Skill, Programming Fundamentals		-
1	**SQL: DDL, DML , DCL statements**	1.1 DDL & Constraints: DDL Statements: Create, Alter, Drop, Truncate, Constraints: Primary key, Unique, Null/Not Null, Foreign Key, and Referential Integrity. 1.2 DML Operations : Insert, Update, Delete 1.3 Views & Access Control: Create View, Grant and Revoke, Grant with grant option, create user, create role. **Self-Learning Topics:** Temporal Tables (System-Versioned Tables) Dynamic Data Masking (DDM) Online reference: https://learn.microsoft.com/en-us/sql/relational-databases/tables/temporal-table-usage-scenarios?view=sql-server-ver17 Dynamic Data Masking - SQL Server | Microsoft Learn	06	LO1,LO2, LO3
2	**SQL: Data Retrieval Statements**	2.1 Basic SELECT & Filtering : Select clause, Where clause, Order by clause, Operators: Between.-And, In, Like 2.2 Aggregate Functions & Grouping: Min, Max, Avg, Sum, Count, Group By, Having 2.3 String & Date Functions: Upper, Lower, Initcap, Concat, Substr, Length, Instr, Trim, Lpad/Rpad, Replace, sysdate, current_date, extract, to_date, to_char, Month_between, add_months, Last_day, Next_day, Trunc 2.4 Joins (Inner, Left, Right, Full) : Cross Join, Inner Join, Natural Join, Outer Join: Left Outer Join, Right Outer Join, Full Outer Join 2.5 Subqueries or Nested Queries : Single Row and Multiple Row subquery, Correlated Subquery , If Exists **Self-Learning Topics:** Window Functions (OVER, PARTITION BY, RANK, LAG/LEAD) Spatial Queries (GIS in SQL) Online reference: https://docs.oracle.com/en/database/other-databases/timesten/22.1/sql-reference/analytic-functions.html#GUID-56F7ECE2-ACA5-4C76-98AA-8B2537D31CE8 https://postgis.net/documentation/	10	LO2, LO3

3	**PL/SQL: Stored Procedure, Functions and Trigger**	3.1 Stored Procedures: Need of stored procedure, Syntax to Create stored procedure, Exec stored procedure. 3.2 Functions: Used-define Functions, Create function, Calling of the function. 3.3 Triggers : Need of trigger, types of trigger, Create trigger , Event-Condition-Action model **Self-Learning Topics:** Error Handling & Exception Propagation INSTEAD OF Triggers on Views Online reference: https://docs.oracle.com/en/database/oracle/oracle-database/26/lnpls/plsql-error-handling.html https://docs.oracle.com/en/database/oracle/oracle-database/26/lnpls/plsql-triggers.html	06	LO4
4	**PL/SQL: Cursor**	4.1 PL/SQL Block & Cursor: Pl/Sql Block format, 4.2 What is a Cursor?, Explicit cursor operations. **Self-Learning Topics:** Parameterized Cursors with Complex Logic Parallel Cursor Execution Online reference: https://docs.oracle.com/en/database/oracle/oracle-database/26/lnpls/static-sql.html https://docs.oracle.com/en/database/oracle/oracle-database/26/dwhsg/parallel-execution.html	02	LO4
5	**NoSql: MongoDB: CRUD and Aggregation Operations**	5.1 MongoDb Vs SQL Key Difference 5.2 CRUD & Document Design: document-based data model, Create, Read, Update, Delete operations, Queries with filters, projections and sorting. 5.3 Aggregation Pipeline: Purpose, Transformations, Grouping, Joins, Array Processing and Performance. **Self-Learning Topics:** MongoDB Realm (Device Sync) Time Series Collections Online reference: https://www.mongodb.com/docs/atlas/device-sdks/deprecation/ https://www.mongodb.com/docs/manual/core/timeseries-collections/	04	LO5
6	**NoSql: MongoDB-Indexing and Replication**	6.1 Indexing & Replication/Sharding (Concept + Basic Ops): Types on Indexes, Analyse index performance, 6.2 Replication: Replica set components: Primary, Secondary, Arbiter, and Quorum. 6.3 Sharding: Shard, Mongos, Shard Key, Config Server. **Self-Learning Topics:** Wildcard Indexes Cross-Region Replication (Global Clusters) Online reference: https://www.mongodb.com/docs/manual/core/indexes/index-types/index-wildcard/ https://www.mongodb.com/docs/atlas/architecture/current/deployment-paradigms/multi-region/ https://oneuptime.com/blog/post/2026-03-31-mongodb-multi-regiondata-distribution/view	02	LO6
Total			**30**	

Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.

References:

1. Arup Nada, Brendan Tierney, Heli Helskyaho ,"SQL and PL/SQL", Singhua University Press , 2019
2. Kristina Chodorow , "MongoDB: The Definitive Guide" (1st & 2nd Editions) O'Reilly Media.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs145
2	Coursera	https://www.coursera.org/fr-FR/learn/implement-polyglot-persistence-using-sql-and-nosql-databases
3	Coursera	https://www.coursera.org/programs/coursera-for-community-guides-neujh/learn/modern-databases-odc

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	To design and create the relational schema for a Healthcare Management System, including tables Patients, Doctors, and Appointments with primary keys, foreign keys, CHECK, NOT NULL, and UNIQUE constraints	2
02	To perform INSERT, UPDATE, and DELETE operations on healthcare records and implement row-level auditing using a separate Audit_Log table that records all DML changes with timestamps and user information.	2
03	To create a restricted view for doctors to see only their own appointments and an admin view with full data, and implement GRANT and REVOKE commands to enforce role-based access control.	2
04	To retrieve patient records based on specific diseases, appointment date ranges, and doctor specializations using WHERE, BETWEEN, LIKE, IN, and ORDER BY clauses.	2
05	To generate summary reports by counting patients per disease, calculating the average age of patients per doctor, and counting total appointments per month using GROUP BY and HAVING.	2
06	To format patient names, extract the year from date of birth, compute current age dynamically, and mask sensitive data (e.g., displaying only the last 4 digits of phone numbers) using Oracle string and date functions.	2
07	To list patients with their appointment details and doctor names by performing INNER, LEFT, RIGHT, and FULL OUTER joins across the Patients, Appointments, and Doctors tables.	2
08	To find patients who have visited more than the average number of times, and solve other nested retrieval problems using IN, EXISTS, ANY, and ALL operators.	2
09	To find patients who have visited more than the average number of times, and solve other nested retrieval problems using IN, EXISTS, ANY, and ALL operators.	2
10	To create a user-defined function that calculates either a patient's visit frequency over a period or the estimated treatment cost based on diagnosis code, and demonstrate its usage in a SELECT query.	2
11	To implement an AFTER DELETE trigger on the Patients table that automatically logs all deleted records (including patient ID, name, deletion time, and database user) into a separate Patient_Audit table for compliance purposes.	2
12	Write a PL/SQL block using explicit cursor to process all patients due for follow-up and send a reminder message.	2
13	Create a patients collection with nested documents (address, insurance, medical history).Insert, query, update, and delete documents using find (), updateOne(), deleteMany().	2
14	Use $match, $group, $unwind, $project to analyze: *Average number of prescriptions per patient, Most common diagnosis by age group*	2

15	Create single and compound indexes (ensureIndex()). – Set up a replica set locally or using MongoDB Atlas. – Enable sharding on a collection using a shard key (e.g., patient_id or hospital_id). – Demonstrate change streams to monitor real-time updates.	2

Suggested List of Assignments:

Sr No	List of Assignments	Hours
01	Q1. Explain the three-layer abstraction architecture of a DBMS. How does it provide logical and physical data independence? Compare this with file systems where such abstraction is absent. Q2. Consider a Healthcare database with entities: PATIENT, DOCTOR, APPOINTMENT, and PRESCRIPTION. Draw an ER diagram showing: - All entities with appropriate attributes (including types: simple, composite, multi-valued, derived). - Relationships with cardinality constraints. - Specialization of PATIENT into INPATIENT and OUTPATIENT with constraints (overlap/cover). - Generalization of DOCTOR and NURSE into STAFF. - A weak entity if applicable. Q3. Define functional dependency and its role in normalization. For the relation APPOINTMENT (PatientID, PatientName, DoctorID, DoctorSpecialty, AppointmentDate, RoomNo) - Identify the candidate key(s). - State the highest normal form (up to 4NF). - Decompose it into 3NF and BCNF (if needed) using decomposition rules. Q4. Explain the stages of query processing. And Differentiate between query parsing and query optimization. Q5. Explain the following integrity constraints with examples from healthcare: - Domain constraint - Entity integrity - Referential integrity (with foreign key) - Aggregation constraint (e.g., a TREATMENT composed of multiple PROCEDUREs) Q6. Write SQL/PLSQL solutions for a hospital database (use schema from Q2): (a) DDL to create PATIENT and VISIT tables with referential integrity and column-level security. (b) A stored procedure that inserts a new visit and updates total bill. (c) A trigger that prevents appointment double-booking. (d) A nested query to find patients with more than 3 emergency visits.	02
02	Q1. A transaction T transfers $500 from Account A to Account B. Explain the ACID properties with respect to this transaction. Also: - Draw the transaction state diagram showing all possible states. - If a system failure occurs after commit, how does recovery work? - If a media failure occurs, explain a recovery technique. Q2. Consider the following schedule of two transactions T1 and T2. Determine if it is conflict serializable. If not, explain why. T1: R(X), W(X), R(Y), W(Y) T2: R(X), W(X), R(Z), W(Z) Also, explain timestamp-based protocol and how it avoids deadlocks. Compare with two-phase locking (2PL). Q3. Explain distributed database architecture types (homogeneous/heterogeneous). Using a multi-hospital system as an example, describe: - Fragmentation (horizontal, vertical, hybrid) with justification. - Replication strategies (synchronous/asynchronous) and trade-offs. - How distributed query processing differs from centralized.	02

	Q4. Compare RDBMS vs NoSQL for a healthcare IoT application (e.g., real-time patient vitals from wearables). Your answer must include: - CAP theorem vs ACID trade-offs. - Explain document, key-value, column-family, and graph NoSQL patterns with use cases. Q5. Write a CRUD operation of MongoDB to insert a patient's vital reading, an aggregation pipeline to compute hourly average heart rate, and explain sharding strategy. Q6. A healthcare provider wants to migrate from on-prem Oracle RDBMS to a cloud solution. Answer: (a) Compare Database-as-a-Service (DBaaS) vs self-managed cloud DB. (b) What is a multi-cloud database strategy? Give pros/cons.	

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	1
2023108	Java Programming Lab	--	--	02	--	--	30	--	30	

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023108	Java Programming Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The healthcare ecosystem in Bharat is increasingly driven by intelligent digital technologies such as Electronic Health Records (EHR), telemedicine platforms, wearable health monitoring systems, and AI-assisted healthcare applications. This laboratory course provides experiential and skill-based learning through the development of healthcare-oriented software using Java and Object-Oriented Programming principles. Aligned with the objectives of NEP 2020, the course emphasizes hands-on learning, problem-solving, critical thinking, innovation, and multidisciplinary application development. Under the theme "Arogya Pranali – Intelligent Health System for Bharat," students will design and implement real-world healthcare solutions, fostering technical competence, creativity, teamwork, and industry-relevant skills. The course contributes to the vision of Digital Bharat while supporting UNSDG 3 (Good Health and Well-being), UNSDG 9 (Industry, Innovation and Infrastructure), and UNSDG 10 (Reduced Inequalities) through the development of accessible, scalable, and technology-enabled healthcare systems.

Lab Objectives:

Sr. No	Lab Objectives
1	To develop practical programming skills in Java using healthcare-oriented applications.
2	To implement object-oriented programming concepts such as classes, objects, inheritance, and polymorphism in intelligent healthcare systems.
3	To design modular and reusable software components for healthcare applications using abstraction and interfaces.
4	To build reliable healthcare software solutions using structured exception handling mechanisms.
5	To implement file handling and collections framework for healthcare data management.
6	To enhance logical thinking, debugging, testing, and software documentation skills through mini healthcare-based applications.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Develop Java programs using basic syntax, control structures, and arrays for healthcare- related problem statements.	L3
2	Design healthcare application models using classes, objects, constructors, and encapsulation techniques.	L3
3	Implement inheritance and polymorphism concepts to create reusable healthcare software components.	L4
4	Develop modular healthcare applications using abstract classes, interfaces, and packages.	L4-L6
5	Construct robust healthcare applications using exception handling mechanisms and user- defined exceptions.	L6
6	Develop healthcare data management applications using Java I/O streams and Collections Framework.	L3/L6

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:
PC's with i3/i5/i7 Processor with at least 16GB of RAM and 512GB/1TB SSD HDD, 21 " LED display.	MS-Windows/Linux Operating System, Latest Java/JDK.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	**Prerequisite**	Fundamentals of C Programming, variables, loops, arrays, functions, and basic problem-solving techniques.	02	-
1	**Java Fundamentals**	1.1. Installation and configuration of JDK and IDE (Eclipse/IntelliJ IDEA/VS Code). 1.2. Writing, compiling, and executing basic Java programs. 1.3. Practical implementation of data types, operators (including unsigned right shift operator), and expressions. 1.4. Control structures: if-else, switch, branching statements. 1.5. Loop constructs: for, while, do-while, and enhanced for-each loop. 1.6. Arrays in Java: Declaration, instantiation, initialization, multidimensional arrays, boundary checking. **Self-Learning Topics:** Digital healthcare applications in Bharat Healthcare data representation using arrays Basics of Electronic Health Records (EHR) Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://abdm.gov.in https://nha.gov.in/NDHM https://www.digitalindia.gov.in/sectors/healthcare	6	LO1
2	**Classes and Objects for Healthcare Applications**	2.1. Classes and Objects: Defining classes, instantiating objects, data members, and member methods. 2.2. The this keyword: Referencing current object context.	6	LO2

		2.3. Methods: Declaration, return types, parameter passing (Pass-by-value in Java). 2.4. Constructors: Default, parameterized, and copy constructors; Constructor overloading and chaining (this() vs super()). 2.5. Access Modifiers: Implementing encapsulation using private, default, protected, and public. 2.6. Static Keyword: Static variables, static methods, and static blocks (class-level vs instance-level memory). 2.7. Strings in Java: String class (immutability), StringBuilder, and StringBuffer classes. 2.8. Input/Output in Java: BufferedReader class, Scanner class. **Self-Learning Topics:** Object-oriented modeling of healthcare systems Secure access control in medical software Healthcare appointment scheduling applications Online reference**:** https://www.javatpoint.info https://www.w3schools.com/java/ https://www.archimetric.com/c4-model-article-healthcare-appointment-booking-system/ https://www.archimetric.com/c4-model-article-healthcare-appointment-booking-system/		
3	**Inheritance and Polymorphism**	3.1. Inheritance: Types of inheritance (Single, Multilevel, Hierarchical); The extends keyword. 3.2. Method Overriding: Redefining behaviors; Rules for overriding. 3.3. Polymorphism: Compile-time polymorphism (Method Overloading) vs Runtime polymorphism (Method Overriding). 3.4. Dynamic Method Dispatch. 3.5. super Keyword: Accessing parent class members and constructors. 3.6. final Keyword: final variables, final methods, and final classes. 3.7. Object Class: Java class hierarchy (toString(), equals(), hashCode()). **Self-learning Topics:** Reusable healthcare software components Generalization and specialization in hospital systems Smart diagnostic service modeling Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://www.geeksforgeeks.org/generalization-specialization-and-aggregation-in-uml/ https://guides.visual-paradigm.com/what-is-generalization-in-uml/	6	LO3
4	**Abstraction, Interfaces, and Packages**	4.1. Abstract Classes: Abstract methods, concrete methods, and architectural use cases. 4.2. Interfaces: Defining pure contracts using interface and implements keywords. 4.3. Multiple Inheritance via Interfaces. 4.4. Default and static methods in interfaces (Java 8+).	4	LO4

		4.5. Packages in Java: package statement, import statement, directory structure, creating user-defined packages. 4.6. The java.lang Package: Automatically imported core utilities. 4.7. Wrapper Classes: Integer, Double, Character, Boolean; Autoboxing and Unboxing. **Self-Learning Topics:** Interoperability in digital healthcare systems Modular healthcare software architecture Telemedicine application design concepts Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://abdm.gov.in/ https://www.healthit.gov/topic/health-it-basics/ https://www.ibm.com/topics/microservices		
5	**Exception Handling**	5.1. The Throwable Hierarchy: Errors vs Exceptions; Checked vs Unchecked (Runtime) exceptions. 5.2. Exception Handling Keywords: try, catch, finally, throw, and throws. 5.3. Multiple Catch Blocks: Catching specific exceptions and the multi-catch block feature. 5.4. Nested try blocks. 5.5. Custom Exceptions: Designing user-defined exception classes for business logic failures. **Self-learning Topics:** Runtime error handling in healthcare systems Medical data validation mechanisms Reliability in intelligent healthcare software Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://www.who.int/health-topics/digital-health https://abdm.gov.in/knowledge-hub	3	LO5
6	**Java I/O Streams and Collections Framework**	6.1. Java I/O Streams: Byte Streams (FileInputStream, FileOutputStream) vs Character Streams (FileReader, FileWriter). 6.2. Buffered Streams: BufferedReader, BufferedWriter for efficient I/O. 6.3. File Handling Basics: Reading from and writing to files. 6.4. The Collections Framework: Why arrays fall short. 6.5. ArrayList: Dynamic arrays for patient management. 6.6. HashMap: Key-value pairs for disease tracking and hospital inventory. 6.7. HashSet: Unique healthcare record storage. 6.8. Iterating Collections using Iterator interface and Enhanced For-Loop. **Self-Learning Topics:** Healthcare data persistence techniques Smart health analytics data structures Patient information management systems Open stalk development	3	LO6

		Online reference: https://www.javatpoint.info https://www.w3schools.com/java/ https://openmrs.atlassian.net/ https://spark.apache.org/ https://www.coursera.org/		
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. H. Schildt and D. Coward, “Java: The Complete Reference”, 13th ed., McGraw Hill Education, 2025.
2. K. Sierra, B. Bates, and T. Gee, “Head First Java”, 3rd ed., O'Reilly Media, 2022.
3. C. S. Horstmann, “Core Java Volume I: Fundamentals”, 13th ed., Pearson (Oracle Press), 2024.
4. E. Balagurusamy, “Object Oriented Programming with Java”, 6th ed., McGraw Hill Education India, 2019.
5. P. J. Deitel and H. M. Deitel, “Java: How to Program”, 11th ed., Pearson, 2018.
6. J. Kvedar, “Digital Health: Connecting and Empowering Patients and Clinicians”, 1st ed., Wiley, 2021.

Online References:

Sr. No.	Website Name	URL
1	Oracle	https://docs.oracle.com/en/java/
2	Open JDK	https://openjdk.org
3	Geeks for Geeks	https://www.geeksforgeeks.org/java/
4	Javatpoint	https://www.javatpoint.info
5	W3school	https://www.w3schools.com/java/

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	Install and configure the Java Development Kit (JDK) and Integrated Development Environment (IDE) on Windows/Linux platforms. Develop and execute basic Java programs.	2
02	Write Java programs using variables, data types, operators, and input/output statements to solve simple computational problems.	2
03	Implement programs using decision-making and looping constructs for solving real-world problems.	2
04	Develop applications using one-dimensional and two-dimensional arrays for data storage and processing.	2
05	Design classes and objects using constructors, methods, and object instantiation.	2
06	Implement encapsulation using private data members, access modifiers, and getter/setter methods.	2
07	Develop applications demonstrating inheritance and code reusability through hierarchical and multilevel class structures.	2
08	Demonstrate polymorphism through method overloading and method overriding.	2
09	Create abstract classes and interfaces to implement abstraction and achieve flexible software design.	2

10	Develop modular applications using packages and access control mechanisms.	2
11	Implement exception handling techniques for robust and reliable program execution.	2
12	Create applications for file creation, reading, writing, and updating using Java File I/O operations.	2
13	Develop applications using collection framework classes such as ArrayList, LinkedList, HashSet, and HashMap.	2
14	Design GUI-based applications using Java Swing/JavaFX components and event handling.	2
15	Build a mini-project integrating object-oriented concepts, exception handling, collections, file handling, and GUI development.	2

Suggested List of Assignments:

Sr No	List of Assignments / Tutorials	Hours
01	Study the architecture of Java (JDK, JRE, JVM) and explain the program compilation and execution process. Compare Java with C/C++ in terms of portability, memory management, and security.	2
02	Analyze the role of decision-making statements, looping constructs, arrays, and strings in problem-solving. Discuss their applications with suitable examples.	2
03	Explain the concepts of classes, objects, constructors, and encapsulation. Illustrate how these concepts contribute to modular and secure software design.	2
04	Prepare a report on the four pillars of Object-Oriented Programming—Encapsulation, Inheritance, Polymorphism, and Abstraction. Discuss their significance in developing reusable and maintainable software systems.	2
05	Study Java packages, access modifiers, and exception handling mechanisms. Explain how they improve code organization, security, and robustness in software applications.	2
06	Investigate the Java Collection Framework and File Handling mechanisms. Compare collections with arrays and discuss the importance of persistent data storage in modern applications.	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2023601	Nirman Lab-III	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2023601	Nirman Lab-III	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Aligned with the National Education Policy (NEP) 2020, the institution emphasizes experiential, interdisciplinary, and project-based learning through the Nirman Lab a central hub for hands-on innovation. To strengthen the undergraduate research ecosystem, the institution has adopted a theme-based academic model aligned with UNSGD. Each semester features six curated problem statements based on local need and aligned with core subjects in the same semester, enabling students to apply classroom knowledge to real-world challenges. Every student selects one problem and develops an individual, subject-integrated solution enhancing both academic understanding and research skills.

The Nirman Lab supports this initiative with facilities for design thinking, prototyping, and product development. Students maintain a project logbook throughout the semester to track their progress and reflections

Lab Objectives:

Sr. No.	Lab Objectives
1	To promote experiential and project-based learning that bridges theoretical knowledge with real-world problem-solving.
2	To encourage interdisciplinary integration by enabling students to apply concepts from multiple subjects within a single cohesive project.
3	To develop innovation and design thinking skills through hands-on activities and iterative solution development.
4	To foster critical thinking and creativity by engaging students in open-ended problems with multiple solution pathways.
5	To enhance communication, collaboration, and documentation skills essential for professional engineering practice.
6	To build an entrepreneurial and research mindset by guiding students to develop scalable socially-relevant, and technically viable prototype.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Recall and articulate key concepts from core and allied subjects relevant to the assigned project	L1
2	Explain the interdisciplinary nature of the problem and the role of each subject in addressing it	L2
3	Apply appropriate tools, techniques, and theoretical knowledge to develop project components	L3
4	Analyze problem constraints and user requirements to structure a feasible and efficient solution	L4
5	Evaluate multiple design options and justify the chosen solution based on technical and practical considerations	L5
6	Create a functional prototype or solution that demonstrates innovation, utility, and integration of interdisciplinary knowledge	L6

Project Guidelines (Interdisciplinary Project Execution in Nirman Lab)

- Each student works on an individual interdisciplinary project aligned with the semester theme.
- Faculty in-charges mentor both the academic and technical aspects, and track weekly progress.
- Project assessment will be rubric-based, ensuring depth, innovation, documentation, and ownership.
- Students shall convert the best solution into working model using various components of their domain areas and demonstrate.

Guidelines for same semester Subject Concepts Applied within the Project

- Term work for each subject will partially reflect how well a student applies subject-specific concepts in their interdisciplinary project.
- Internal assessment panel will collaborate to align project components with subject learning outcomes.

Role of Faculty In-Charges in Nirman Lab Projects

Faculty in-charges play a pivotal role in the success of interdisciplinary, theme-based projects under the Nirman Lab. Their responsibilities extend beyond technical supervision to include academic alignment, innovation facilitation, and active student engagement. Their key roles include:

- Motivating and Inspiring Students
  - Encourage students to take ownership of their learning and projects.
  - Cultivate a mindset of curiosity, exploration, and social relevance.
  - Foster an environment where students feel empowered to take creative risks.
- Conducting Brainstorming and Ideation Sessions
  - Organize structured brainstorming sessions at the start of the semester to help students define their problem statements and solution pathways.
  - Promote collaborative thinking, design exploration, and interdisciplinary integration.
- Arranging Guest Lectures and Expert Talks
  - Identify and invite industry experts, researchers, and innovators for guest lectures aligned with the semester's theme or subject areas.
  - Facilitate exposure to real-world challenges, current trends, and future opportunities.
- Ensuring Uniqueness and Originality of Projects

    - Actively review proposed ideas to ensure **no duplication of solutions** across students.
    - Encourage students to explore novel approaches, technologies, and perspectives.
- Promoting Discussion and Collaborative Learning
    - Create platforms for students to present, discuss, and receive peer and mentor feedback.
    - Facilitate idea refinement through regular discussions and group engagement.
- Monitoring and Documentation
    - Oversee project logbook maintenance, milestone tracking, and submission of progress reports.
    - Provide ongoing feedback and ensure project alignment with learning outcomes.
- Coordination with Subject Faculty
    - Work in collaboration with other subject faculty to help students embed theoretical and practical aspects of their coursework into the project.
    - Facilitate subject-term mapping and contribute to termwork assessment based on evidence.

Guidelines for Assessment

Presentation-Based Assessment Structure (Total: 25 Marks)

Assessment Month	**Weightage**	**Marks**
Month 1 (Formative 1)	20%	5 marks
Month 2 (Formative 2)	40%	10 marks
Month 3 (Formative 3)	40%	10 marks

Rubric-Based Evaluation Criteria

Criteria	**Month 1 (5)**	**Month 2 (10)**	**Month 3 (10)**
Problem Understanding	Connects problem to subjects	Defines interdisciplinary scope	Demonstrates deep conceptual grasp
Subject Knowledge Application	Identifies relevant concepts	Applies principles in design	Integrates multiple subject areas correctly
Innovation & Design Thinking	Proposes creative idea	Develops and tests feasible solution	Final solution shows originality and utility
Documentation & Presentation	Logbook initiated, plan presented	Mid-design log and visuals	Final report and demo completed
Progress & Ownership	Meets deadlines, shows planning	Demonstrates self-motivation	Completes project independently with reflection

Total Assessment Structure

Component	**Marks**	**Assessed By**
Termwork – Project Execution	25 Marks	Project Guide
Viva Voce (Final Evaluation)	25 Marks	External Examiner

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2993501	Entrepreneurship Development	--	-	2*+2			60	-	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2993501	Entrepreneurship Development	-	-	-	-	-	50		50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The subject Entrepreneurship and Business Model Development aligns strongly with the vision of the National Education Policy (NEP) 2020, which emphasizes experiential learning, multidisciplinary education, and the cultivation of innovation and problem-solving skills. This course nurtures an entrepreneurial mindset among students by integrating theoretical concepts with practical applications such as ideation, prototyping, and business model design. It encourages critical thinking, creativity, and risk-taking abilities, which are essential for self-reliance and employability. By incorporating project-based learning, case studies, and industry interaction, the subject supports NEP 2020's focus on skill development, vocational integration, and fostering job creators rather than job seekers.

Furthermore, the course contributes significantly to the achievement of United Nations Sustainable Development Goals (UNSDGs) by encouraging students to design sustainable, inclusive, and socially responsible business models. It promotes awareness of global challenges such as poverty, inequality, climate change, and responsible consumption, aligning particularly with SDG 4, 8, and 9.

Lab Objectives:

Sr. No.	Lab Objectives
1	To introduce students to the fundamental concepts, theories, and types of entrepreneurship, and develop their ability to analyze entrepreneurial opportunities using tools like SWOT.
2	To introduce students to the concepts, processes, and methods of innovation, enabling them to generate ideas and develop start-up ventures using design thinking while understanding legal and IP considerations.
3	To enable students to understand and apply the concepts of new product development, feasibility analysis, test marketing, and commercialization strategies for successful product launch.
4	To familiarize students with various funding sources and legal frameworks involved in establishing and protecting a business venture.
5	To develop the ability to design effective business models and evaluate entrepreneurial opportunities through feasibility, industry, and competitor analysis.
6	To enable students to understand funding mechanisms, government support schemes, and the process of preparing and presenting a comprehensive business plan.

Lab Outcomes:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Students will be able to understand and analyse different forms and theories of entrepreneurship and evaluate entrepreneurial opportunities using SWOT analysis and case studies.	L2,L4,L5
2	Students will be able to apply innovation theories, systems thinking, and design methodologies to develop and evaluate start-up ideas with awareness of legal and intellectual property issues.	L3,L5
3	Students will be able to generate, evaluate, and commercialize new product ideas using appropriate development, testing, and market entry techniques.	L5,L6
4	Students will be able to identify suitable funding options and apply basic legal and IPR procedures for starting and safeguarding a business.	L1,L3
5	Apply business model frameworks such as the Business Model Canvas to identify opportunities and analyse the feasibility and competitiveness of entrepreneurial ideas.	L3,L4
6	Apply knowledge of venture financing, business planning, and entrepreneurial support systems to develop and effectively pitch a viable business plan.	L3,L6

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mapping
0	**Prerequisite**	Introduction to Entrepreneurship and Business	00	
1	**Introduction to Entrepreneurship**	1.1 Concept of Entrepreneurship - Meaning, definition and overview of Entrepreneurship, Evolution and growth of entrepreneurship in India, Characteristics of entrepreneurs, types of entrepreneurs. 1.2 Theories of entrepreneurship- Entrepreneurship, Women Entrepreneurship, Social entrepreneurship, SWOT Analysis, case study. **Self-learning topic**: SWOT analysis case study of any company Online Reference: https://www.classcentral.com/classroom/youtube-innovation-business-models-and-entrepreneurship-47483 https://ocw.tudelft.nl/course-lectures/2-1-makes-good-business-model/	10	LO1
2	**Innovation**	2.1 Evolution and theories of Innovation- Meaning, concept, Definition. The Process of Innovation and Idea Generation, Social innovation 2.2 Method of Innovation- Systems Thinking as a Method for Innovation. Team Formation. Problem Definition and Asset Mapping. Development of Start-up Venture Idea, design thinking and innovation. Legal And IP issues in innovation, case study. **Self learning topic**: different start ups in India Online Reference: https://www.classcentral.com/classroom/youtube-innovation-business-models-and-entrepreneurship-47483 https://cursa.app/en/free-course/bussiness-models-and-entrepreneurship-diib	10	LO2
3	**Product Development**	3.1 Concept of new product development- idea generation, sources of new ideas, methods of idea generation, Feasibility study. 3.2 Marketing and commercialization - Test marketing- Advantages, need, types, and methods. Prototype and MVP. Commercialization- Time to market, Breaking into the market, methods of commercialization **Self learning topic**: Marketing strategies case study Online Reference: https://www.classcentral.com/classroom/youtube-innovation-business-models-and-entrepreneurship-47483 https://cursa.app/en/free-course/bussiness-models-and-entrepreneurship-diib	10	LO3

4	**Funding and Legal Framework**	4.1 Sources of Funding- Bootstrapping, Angel Investors, Venture Capital Government Schemes & Start-up India Initiatives. 4.2 Business Registration & Legal Formalities- Intellectual Property Rights (IPR) & Patents **Self-learning topic**: IPR and Patent rules in India Online Reference: https://www.classcentral.com/classroom/youtube-innovation-business-models-and-entrepreneurship-47483	10	LO4
5	**Business Models**	5.1 Different Business Models- Developing and effective business model, entrepreneurship models, corporate entrepreneurship models. Elements of business models, The business model canvas 5.2 Opportunities and Challenges- Recognizing opportunities and generating ideas, Feasibility analysis, Industry, and competitor analysis, Case study. **Self-learning topic:** Challenges in different domains in IT **Online Reference**: https://cursa.app/en/free-course/business-model-canvas-course-strategyzer-framework-and-startup-strategy-feif https://ocw.tudelft.nl/course-lectures/2-1-makes-good-business-model/	10	LO5
6	**Financing and preparing a business plan**	**6**.1 Financing and Business- Financing the venture, Sources of Finance, Bootstrapping strategies and their Valuation, Crowd-funding, and its valuation. Role of government and Other agencies in entrepreneurship. 6.2 Business Plan- Entrepreneurial assistance schemes, the business plan- Contents, preparing a business plan, pitching, Case study **Self-learning topic**: Finance schemes in India Online Reference: https://cursa.app/en/free-course/bussiness-models-and-entrepreneurship-diib https://www.ocw.mit.edu/courses/15-390-new-enterprises-spring-2013/resources/mit15_390s13_lec14/	10	LO6
		Total	**60**	

Note:

1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Amit, R., & Zott, C. (2020). Business model innovation strategy: Transformational concepts and tools for entrepreneurial leaders. John Wiley & Sons.

2. Lewrick, M. (2022). Design Thinking for Business Growth: How to Design and Scale Business Models and Business Ecosystems. John Wiley & Sons.
3. Mitra, J. (2019). Entrepreneurship, innovation, and regional development: an introduction. Routledge.
4. Lorenzo, O., Kawalek, P., & Wharton, L. (2018). Entrepreneurship, Innovation, and technology: A guide to core models and tools. Routledge.

Online References:

Sr. No.	Website Name	URL
1	Sustainable entrepreneurship, innovation, and business models: Integrative framework and propositions for future research - Lüdeke-Freund – 2020, Business Strategy and the Environment - Wiley Online Library	https://onlinelibrary.wiley.com/doi/full/10.1002/bse.2396
2	Youtube	https://www.youtube.com/watch?v=rcGcpd03Bjk&pp=ygUwRW50cmVwcmVuZXVyc2hpcCwgSW5ub3ZhdGlvbiBhbmQgQnVzaW5lc3MgTW9kZWxz
3	Innovation, Business Models and Entrepreneurship - Course (nptel.ac.in)	https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_mg63
4	Entrepreneurship and Innovation Toolkit (bccampus.ca)	https://pressbooks.bccampus.ca/entrepreneurship/chapter/chapter-4-business-models/

Suggested List of Experiments:

Sr. No.	List of Experiments	Hours
1	Study and Application of SWOT Analysis for Entrepreneurial Evaluation	2
2	Analysis of Entrepreneurial Types and Theories through Case Study	3
3	Study and Application of Innovation Process for Startup Idea Generation	2
4	Applying Design Thinking and Systems Thinking for Startup Idea Development with Legal and IP Considerations	3
5	Idea Generation and Feasibility Analysis for New Product Development	2
6	Prototype/MVP Development and Test Marketing Strategy for Product Commercialization	3
7	Study and Analysis of Startup Funding Sources	2
8	Study of Business Registration Process and Intellectual Property Rights (IPR)	2
9	Design and Analysis of a Startup using Business Model Canvas	3
10	Identification of Business Opportunities and Feasibility Study	3
11	Study and Evaluation of Different Sources of Finance and Bootstrapping Strategies	2
12	Preparation of a Business Plan and Pitch Deck for a Startup Idea	3

Suggested List of Assignment:

Sr. No.	List of Assignment	Hours
1	Describe the evolution and growth of entrepreneurship in India. Highlight key phases and government initiatives.	2
2	Propose a startup idea based on a real-world problem and Apply design thinking steps (Empathize, Define, Ideate, Prototype, Test) to your idea.	2
3	Case Study: Select a startup/product (e.g., Swiggy, Zomato, Ola, or any tech product) and analyze: Idea origin MVP or prototype used Marketing strategy Commercialization approach	2
4	A group of IT engineering students plans to launch a startup focused on developing an AI-based traffic management system. Answer the following: Funding Strategy Legal & Registration Requirements	2
5	Identify a real-world problem and propose an innovative solution. Justify why the idea is viable	2
6	Conceptualize a startup idea and prepare a comprehensive report covering financing strategies, valuation approaches, and a detailed business plan.	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 50 Marks (Total) = 25 Marks (Experiments) + 10 Marks (Assignments) + 05 Marks (Attendance) + 10 (Presentation)

vCourse Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2993502	Engineering for Sustainability	--	--	2*+2	--	--	60	--	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2993502	Engineering for Sustainability	---	---	---	---	---	50		50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Most engineering branches originate from applied sciences and significantly impact the environment. Aligned with NEP 2020's approach and theme, understanding environmental studies is essential for engineers to develop sustainable solutions, minimize ecological footprints, and promote responsible resource management. This course integrates ecosystems, biodiversity, pollution control, and environmental laws, directly supporting UNSDG goals such as SDG 2 (Zero Hunger), SDG 6 (Clean Water), SDG 12 (Responsible Consumption), SDG 13 (Climate Action), and SDG 15 (Life on Land), enabling engineers to embed sustainability into practice.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand the scope, importance, and role of environmental studies in public awareness and health.
2	To study different natural resources, their issues, and sustainable conservation.
3	To understand ecosystem types, structures, and functions.
4	To explore biodiversity, its importance, threats, and conservation.
5	To learn about pollution types, causes, effects, and control measures.
6	To understand environmental challenges, sustainability, and ethics.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the significance of environmental studies and the role of IT in environment and health.	L2
2	Describe resource types, associated problems, and conservation methods.	L4
3	Classify ecosystems and explain their role in ecological balance	L4
4	Analyze biodiversity levels and conservation strategies, especially in India.	L4

5	Explain pollution impacts and suggest preventive measures.	L5 , L6
6	Discuss environmental issues and propose sustainable solutions.	L6

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	**Prerequisite**	Awareness of natural resources, ecosystems, and environmental issues affecting society, Basic understanding of science concepts studied at the secondary and higher secondary level (Physics, Chemistry, Biology, and Geography).	02	-
1	**The Multidisciplinary Nature of Environmental Studies**	1.1 Definition, scope and importance. Need for public awareness, Role of information technology in environment and human health. 1.2 Human population and the environment: Population growth, variation among nations. 1.3 Population Explosion- family welfare program. Environment and human health Women and child welfare **Self-Learning Topics:** Green Computing and Sustainable IT Practices , Role of AI and IoT in Environmental Protection Online References: https://www.coursera.org/specializations/greenskills https://www.udemy.com/course/green-it-professional-sustainable-it-practices/	08	LO1
2	**Natural Resources**	2.1 Renewable and non-renewable resources. Natural resources & associated problems: a) Forest resources: Water resources: Natural resources & associated problems c) Mineral resources: d) Food resources: Energy resources: Role of an individual in conservation of natural resources: Equitable use of resources for sustainable lifestyles. **Self-Learning Topics:** Water Conservation Techniques in Smart Cities , Renewable Energy Technologies (Solar, Wind, Biogas) Online References**:** https://www.coursera.org/learn/sustainable-urban-water-systems https://ieeexplore.ieee.org/document/10370989/	10	LO2
3	**Ecosystems**	3.1 Concepts of an ecosystem. Introduction, types, characteristic features, 3.2 structure and function of the following ecosystem: a. Forest ecosystem b. Grassland ecosystem	10	LO3

		c. Desert ecosystem d. Aquatic ecosystem (ponds, streams, lakes, rivers, oceans, estuaries). 3.3 Case study on various ecosystems in India. **Self-Learning Topics:** Smart Ecosystem Monitoring using Sensors and IoT , Wetland and Mangrove Ecosystems in India Online References: https://www.semtech.com/applications/internet-of-things/smart-environment https://www.coursera.org/specializations/wetland-science-and-conservation		
4	**Biodiversity and its Conservation**	4.1 Introduction, Definition, genetic species and ecosystem diversity. 4.2 Bio-geographical classification of India Value of biodiversity, 4.3 Consumptive use, productive use, social, ethical, aesthetic and option values, 4.4 Bio-diversity at global, national, local levels India as a mega diversity nation, Case study on Bio diversity in India. **Self-Learning Topics:** Biodiversity Hotspots of India , GIS and Remote Sensing Applications in Biodiversity Conservation Online References: https://www.pmfias.com/biodiversity-hotspots-india/ https://www.researchgate.net/publication/270453897_Remote_Sensing_and_GIS_for_Biodiversity_Conservation	10	LO4
5	**Environmental Pollution Definition**	5.1 Causes, effects and control measures of: a) Air pollution b) Water pollution c) Soil pollution. 5.2 Solid waste management: Causes, effect and control measures of urban and industrial wastes. 5.3 Role of an individual in prevention of pollution, 5.4 Case study on Pollution Disaster management: floods, earthquake, cyclone and landslides. Carbon Credits for pollution prevention. **Self-Learning Topics:** E-Waste Management and Recycling Technologies , Carbon Footprint and Carbon Credit Mechanisms Online References: https://www.sciencedirect.com/science/article/pii/S2214993724003622 https://climateseed.com/blog/understanding-carbon-credits	10	LO5
6	**Social Issues and Environme**	6.1Unsustainable to sustainable development Urban problems related to energy, Water conservation, rain water harvesting, watershed management.	10	LO6

	nt	6.2 Environmental ethics: issues and possible solution. 6.3 Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case studies. Consumerism and waste products. Environment protection act. Public awareness Case study on Environmental Ethics **Self-Learning Topics:** Sustainable Development Goals (SDGs) and Environmental Sustainability , Smart Cities and Sustainable Urban Development Online References: https://www.undp.org/sustainable-development-goals https://link.springer.com/book/10.1007/978-3-031-99001-4		
Total			**60**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Environmental Science: Towards a Sustainable Future, G. Tyler Miller and Scott Spoolman, 13th Edition, Cengage Learning 2021
2. Environmental Management: Text and Cases, Bala Krishnamoorthy, 3rd Edition, PHI Learning, Publication Year: 2016
3. Green IT: Concepts, Technologies, and Best Practices, Markus Allemann, Springer 2008
4. Sustainable IT: Slimming Down and Greening Up Your IT Infrastructure, David F. Linthicum, IBM Press 2009
5. Environmental Modelling: Finding Solutions to Environmental Problems, David L. Murray, Cambridge University Press 2016
6. Remote Sensing and Image Interpretation, Thomas M. Lillesand, Ralph W. Kiefer, and Jonathan W. Chipman, 9th Edition, John Wiley & Sons 2020
7. Business Ethics: Concepts and Cases, Manuel Velasquez, Pearson 2012
8. Environmental Law and Policy in India, Shyam Divan and Armin Rosencranz, 2nd Edition, Oxford University Press 2018

Online References:

Sr. No.	Website Name	URL
1	Centre for Science and Environment (CSE)	cseindia.org
2	Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India	https://moef.gov.in/
3	CSIR-National Environmental Engineering Research Institute (NEERI)	https://www.neeri.res.in/

Suggested List of Experiments:

Sr No	List of Experiments	Hours

01	Study of Environmental Components and Ecosystems.	2
02	Visit and Report on Solid Waste Management Plant.	2
03	Study of Renewable Energy Sources (Solar, Wind, Biogas).	2
04	Analysis of Air and Water Quality Parameters.	2
05	Study of Local Biodiversity and Conservation Methods.	2
06	Awareness Activity on Environmental Issues.	2
07	Rainwater Harvesting System Design	2
08	Case Study on Environmental Pollution & Control Measures.	2
09	Report on Climate Change Impact and Adaptation.	2
10	Study of Environmental Laws and Acts.	2
11	Study of Disaster Management Techniques.	2
12	Report on Role of IT in Environmental Protection.	2

Suggested List of Assignment:

Sr No	List of Assignments / Tutorials	Hours
01	Prepare a report on Renewable and Non-Renewable Resources.	2
02	Write a case study on Ecosystem Types in India	2
03	Write a report on Biodiversity in India.	2
04	Prepare a report on Pollution Types and Control Measures.	2
05	Prepare a report on Environmental Ethics and Sustainability.	2
06	Prepare a case study report on Global Warming and Climate Change.	2
07	Report on Role of an Individual in Environmental Protection.	2
08	Write a report on Disaster Management Techniques.	2
09	Prepare a report on Environmental Laws and Acts in India.	2
10	Case Study on E-waste Management and Recycling Techniques.	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 50 Marks (Total) = 25 Marks (Experiments) + 10 Marks (Assignments) + 05 Marks (Attendance) + 10 (Presentation)

S.E. Semester – IV

Bachelor of Engineering Computer Engineering

Syllabus

Theme
Arambh: From Prototype to Product

Aligned with UNSDG: · SDG 9

Keywords: Prototype, Product Development, Startup, Entrepreneurship, Innovation, MVP, Market Readiness, Intellectual Property, Design Thinking, Scalability, Technology Commercialisation, Incubation, Funding, Go-to-Market

Theme Description

The theme **"Arambh: From Prototype to Product"** is a transformative framework that bridges the critical gap between a working prototype and a market-ready product. It recognises that innovation does not end at the laboratory bench or the college hackathon it truly begins the moment an idea must withstand the rigours of real-world users, market constraints, regulatory environments, and scalable engineering. Arambh is a call to engineering students, innovators, and aspiring entrepreneurs to take their validated ideas forward with purpose, discipline, and commercial intent.

In the Indian context, there exists a significant "valley of death" between proof-of-concept demonstrations and deployable products. Thousands of student projects and lab prototypes are built every year, yet very few make the transition to products that reach actual users. **Arambh** addresses this gap head-on challenging participants to think beyond the demo, to ask not just "does it work?" but "who needs it, how will it reach them, and how will it sustain itself?"

The theme spans the full product development lifecycle: from **defining the problem and validating the user,** through **building a Minimum Viable Product (MVP),** to **iterating based on feedback, protecting intellectual property, planning go-to-market strategies, and designing for scale.** It equally embraces the ecosystem that makes product success possible incubators, angel investors, government schemes such as Startup India, DPIIT recognition, and technology transfer frameworks.

Arambh aligns firmly with India's national innovation agenda the **National Innovation and Startup Policy,** NITI Aayog's Atal Innovation Mission, and the **Make in India** and **Atmanirbhar Bharat** missions which collectively envision India as a global hub for product innovation and technology entrepreneurship. The theme invites participants to contribute directly to this vision by transforming their engineering education into market impact.

Ultimately, Arambh is not just about building products it is about building **product thinkers:** engineers who understand users, designers who think about business, and entrepreneurs who execute with technical precision. It is a theme that sees every prototype not as an endpoint but as the **beginning the Arambh of something greater.**

Alignment with UNSDG 9: Industry, Innovation & Infrastructure

SDG 9 calls to "build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation." Arambh is perhaps most directly aligned with this goal, as its entire premise is the operationalisation of innovation into industry-grade products:

- **Fostering Technological Innovation and Commercialisation:** Arambh explicitly targets the gap between innovation and commercialisation the stage where most Indian student projects stall. By building skills in MVP development, user testing, product-market fit validation, and technology transfer, the theme directly advances SDG 9.5's target of enhancing scientific research, upgrading technological capabilities, and encouraging innovation.
- **Building Resilient and Scalable Digital Infrastructure:** Products developed under Arambh spanning IoT, AI, embedded systems, cloud platforms, and mobile applications contribute to India's growing digital infrastructure. Emphasis on scalable architecture, security, and interoperability ensures that student-built products can grow into infrastructure-grade solutions, aligning with SDG 9.1's target of developing reliable, sustainable, and resilient infrastructure.

- **Inclusive Industrialisation through Bharat-Focused Products:** The theme strongly encourages products designed for Tier 2, Tier 3, and rural India solutions that address the real problems of farmers, artisans, small businesses, and underserved communities. This inclusive product philosophy advances SDG 9.2's target of promoting inclusive and sustainable industrialisation and significantly raising industry's share of employment and GDP in developing contexts.
- **Strengthening the Innovation Ecosystem:** Arambh exposes students to the full innovation ecosystem incubators, accelerators, angel networks, government schemes (Startup India, AIM, TIDE 2.0), and technology business incubators (TBIs). This systemic literacy contributes to SDG 9.b's target of supporting domestic technology development, research, and innovation in developing countries.

Problem Statement

1. Navi Mumbai generates large quantities of residential, commercial, and industrial waste, while source segregation and recycling compliance remain inconsistent.

2. Thousands of workers employed in MIDC areas, construction projects, logistics hubs, and port operations face challenges in accessing preventive healthcare and regular medical monitoring.

3. Large residential complexes and commercial buildings consume substantial energy, often without real-time monitoring or optimization mechanisms.

4. Public transportation systems, railway stations, hospitals, and government facilities still present accessibility challenges for persons with disabilities.

5. Major corridors such as Palm Beach Road, Sion-Panvel Highway, Thane-Belapur Road, and node intersections experience recurring congestion.

6. Water leakage, uneven distribution, and increasing demand create challenges for sustainable water management across rapidly growing urban areas.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2024101	Theory of Computation	02	--	--	30	--	--	30	90	3
2024104	Theory of Computation Tutorial	--	01	--	--	15	--	15		

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024101	Theory of Computation	20	20	40	60	2	25	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning
Note: Same Syllabus is prescribed for both theory and Practicals

Rationale:

Theory of Computer Science (TCS) provides the mathematical foundation of computing automata, formal grammars, and computability that underlies every software product and intelligent system. Aligned with the theme "Arambh: From Prototype to Product", TCS serves as the theoretical prototype upon which reliable, scalable computing products are engineered.

Consistent with NEP 2020, the subject fosters analytical thinking, conceptual depth, and the ability to connect formal theory with real-world problem solving. Alignment with UN SDG 9 Industry, Innovation and Infrastructure is reflected in how mastery of computational models empowers students to design efficient, innovative, and sustainable technology solutions that drive industrial growth.

Course Objectives:

Sr. No.	Course Objectives
1	Understand formal models of computation and their applications in Computer Engineering.
2	To develop the ability to construct regular expressions and build equivalent automata models for language representation..
3	To equip students to devise regular and context-free grammars for generating and recognizing strings and tokens.
4	To enable students to design pushdown automata and apply them for recognizing context-free languages.
5	Analyze computational problems using Turing Machines and computability theory.
6	Understand the concept of Undecidability.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Design and analyze finite automata for recognizing formal languages.	L4,L6
2	Construct regular expressions and equivalent automata models.	L6
3	Devise regular, context free grammars while recognizing the strings and tokens.	L6

4	Design pushdown automata to recognize the language.	L6
5	Analyze Turing Machines and computational models.	L4
6	Evaluate decidability and undecidability of computational problems.	L5

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Sets, Relations, Functions, Logic and Graph Concept.	-	-
1	**Basic Concepts and Finite Automata**	1.1 Alphabets, Strings, Languages, Operations on Languages, Grammar and Language Hierarchy 1.2 Deterministic Finite Automata (DFA) and Nondeterministic Finite Automata (NFA): Definitions, transition diagrams and Language Recognizers, NFA to DFA Conversion, Equivalence between NFA with and without ε-Transition 1.3 FSM with output: Moore and Mealy machines, Equivalence, Applications of FA in NLP **Self-Learning Topics:** Automata applications in search engines and pattern matching. Online reference: https://nptel.ac.in/courses/106106049 https://www.youtube.com/watch?v=al4AK6ruRek&t=17s https://www.youtube.com/watch?v=_9fuEO5khrI&t=65s	07	CO1
2	**Regular Expressions and Languages**	2.1 Regular Expressions, Equivalence of RE and FA, Arden's Theorem 2.2 Pumping Lemma of RL, Closure Properties of RL **Self-Learning Topics:** Regex engines used in Python and cybersecurity tools. Application of Regular Languages in Compiler Design, Text Processing Online reference: https://www.youtube.com/watch?v=vQeROGcwQs4&t=3s https://www.youtube.com/watch?v=NfhXY_nttSg	05	CO2
3	**Context-Free Grammars**	3.1 Chomsky hierarchy, CFLs 3.2 Definition, Sentential forms, Leftmost and Rightmost derivations, Parse tree, Ambiguity, Simplification 3.3 Normal Forms: Chomsky Normal Forms (CNF) and Greibach Normal Forms (GNF). Closure properties,Application **Self-Learning Topics:** ANTLR and parser generators. Left Recursion and Its Elimination, Applications of CFGs in XML Parsing, and Natural Language Processing (NLP). Online reference: https://nptel.ac.in/courses/106106049 https://nptel.ac.in/courses/106104028 https://www.youtube.com/watch?v=MTIHtZ633Tc&t=4s https://www.youtube.com/watch?v=-aIRqNnUvEg	06	CO3

4	**Pushdown Automata(**	4.1 Definition, Language of PDA,PDA as generator, decider and acceptor of CFG. 4.2 Deterministic PDA , Non-Deterministic PDA, Equivalence of PDA and CFG, Application of PDA. **Self-Learning Topics:** Parsing & PDA: Top-Down Parsing, Bottom-up Parsing, Closure properties and Deterministic PDA. Online reference: https://nptel.ac.in/courses/106106049 https://www.youtube.com/watch?v=_3Fi5jBhinE https://www.youtube.com/watch?v=8O6FUFwWt5A	05	CO4
5	**Turing Machine**	5.1 Definition, Design of TM as generator, decider and acceptor, Variants of TM, Universal TM, Applications, and Limitations of TMs. **Self-Learning Topics:** Quantum Computing vs Turing Computation. Algorithms using Turing Machine, The Model of Linear Bounded Automata Online reference: https://nptel.ac.in/courses/106106049 https://www.youtube.com/watch?v=iGv2OjgKOe8 https://www.youtube.com/watch?v=IhyEGNn-7Uo	05	CO5
6	**Decidability and Undecidability**	6.1 Decidability and Undecidability, Recursive and Recursively Enumerable Language, 6.2 Halting Problem, Rice's Theorem, Post Correspondence Problem. . **Self-Learning Topics:** NP Completeness of the SAT Problem, A Restricted Satisfiability Problem Online reference: https://nptel.ac.in/courses/106106049 https://www.youtube.com/watch?v=q_DngTMHtV0 https://www.youtube.com/watch?v=KJD_F-QGLmo	02	CO6
		Total Hours	**30**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

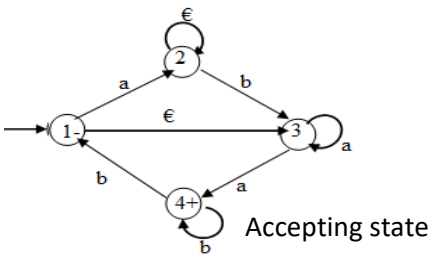
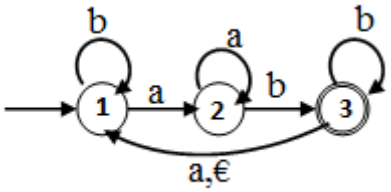
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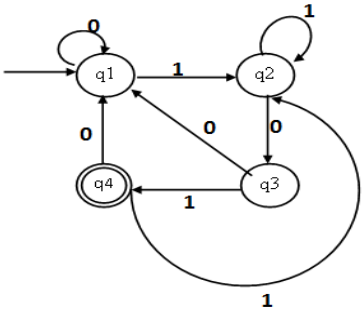
1. John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman, Introduction to Automata Theory, Languages and Computation, Pearson Education, 3rd Edition / 2008.
2. Kavi Mahesh, "Theory of Computation: A Problem Solving Approach", Wiley-India.
3. Vivek Kulkarni, "Theory of Computation", Oxford University Press, India.
4. John C. Martin, Introduction to Languages and the Theory of Computation, McGraw-Hill, 4th Edition / 2010.
5. K. L. P. Mishra and N. Chandrasekaran, Theory of Computer Science, PHI Learning, 3rd Edition / 2012.

Online References:

Sr. No.	Website Name	URL
1	NPTEL Swayam	https://nptel.ac.in/courses/106106049
2	NPTEL Swayam	https://nptel.ac.in/courses/106104028

Suggested List of Tutorials:

Sr No	List of Tutorial	Hours
01	Designing FSM 1. Design FSM in which input is valid if it ends in 'bab' over $\sum$ ={a,b} 2. Design FSM to implement binary adder over $\sum$ ={0,1} 3. Design FSM to recognize a language in which every a's followed by b's over $\sum$ ={a,b} 4. Design FSM in which input is valid if it does not contains 'bbb' over $\sum$ ={a,b} . 5. Convert the full NFA with ε to a complete DFA.  6. Design Mealy machine to find out 2's complement of a binary number. 7. Design Moore machine for the language (0+1)*(00+11) and convert it to Mealy machine. 8. Design Mealy m/c to o/p even & odd depending on the no. of 1's encounters in the i/p string over $\sum$ = {0, 1} and minimize Mealy m/c.	4
02	1. Write a regular expression for the language in which first character is a or c followed by any string in b over Σ = {a, b, c}. 2. Write a regular expression for strings containing no double letter over Σ = {a, b}. 3. Give the regular expression for the following i] Set of all string over {0,1}that ends with 1 and has no substring 00. ii] Set of all string over {0,1}with even no of 1's followed by odd no of 0's. 4. Construct the DFA that accepts the language represented by a) 0*1*2* b) (11+01)* c)  5. Give application of Regular Expression and Finite Automata 6. State and explain pumping lemma for regular languages. 7. Prove that the following languages are not regular i) L= {ww | w∈ (0,1)*} ii) L= {0^i1^i | i>=1} iii) **L={W c W^R | W ∈** (0+1)*}is not regular where w^R is reverse of w.	2

	8. Find R.E. equivalent to following FA.  **9.** Find R.E. for following: Set of all strings over {1,0} that ends with 1 and has no substring 00.	
03	1. What is ambiguous grammar ?Give one example Let G be the grammar S→ aB | bA A→ a |aS| bAA B→ b |bS| aBB Find leftmost derivation, rightmost derivation and parse tree for the string **"bbaaabbaba" .** 0. The following grammar becomes S □A|B A □0A|є B □0B|1B|є Give leftmost and rightmost derivations of the following strings a) 00101 b) 1001 c) 00011 0. Consider the grammar S → aS|aSbS|є Show that derivation for the string aab is ambiguous 0. Consider the grammar where P= S□aAS | a A□SbA | SS |ba Perform LMD,RMD and construct Parse Tree for string**'aabbaa'**	3
04	1. Design PDA for 1] $L=\{a^{n-1} b^{2n+1} \mid n>=1\}$ 2] $L=\{a^{2n} b^{n} \mid n>o\}$ **2. 1] Design PDA to check even parenthesis over ∑={0,1}** 2] Design DPDA for L={ x | Na(x) > Nb(x)} 3. Generate Rule for the grammar **E→E+E | E-E | (E) | id** 4. Give application of PDA 5.a) Design PDA for language $L=\{WCW^R \mid W \in (a,b)^*\}$where W^R is reverse of w b) Compare DFA and PDA	2
05	1. Design TM accepting all palindromes over{0,1} 2. Design Turing machine for well-formed parenthesis. 3. Design TM for the regular expression 0(0+1)*11 4. Give application of Turing machine. 5. Discuss variants of Turing machine. 6. Write short note on universal turing machine 7. Design Turing machine to decrement binary number	3

06	1. Explain undecidability and incompleteness of TM. 2. State and prove Rice's theorem 3. State and prove halting problem 4. Explain PCP along with suitable example	1

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2024102	Operating System	03	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024102	Operating System	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The Operating System course introduces students to the fundamental principles of resource management and system software that act as an interface between computer hardware and users. The course covers essential concepts such as process management, CPU scheduling, memory management, synchronization, deadlocks, file systems, device management, virtualization, and modern operating systems. Through practical and experiential learning, students gain system-level problem-solving skills required for efficient software execution and secure computing environments. Aligned with the Semester IV theme "Arambh: From Prototype to Product," the course enables students to optimize and scale software prototypes into efficient, secure, and deployable digital products. In line with National Education Policy 2020 (NEP 2020), the course promotes skill-based learning, critical thinking, self-learning, and industry readiness through hands-on activities and contemporary topics. It also supports relevant United Nations Sustainable Development Goals (UNSDGs), 4, 8, and 9.

Course Objectives:

Sr. No.	Course Objectives
1	Understand the architecture and functions of operating systems.
2	Apply process management and CPU scheduling algorithms to solve resource allocation problems.
3	Analyze synchronization mechanisms and deadlock handling methods.
4	Evaluate memory management techniques and virtual memory schemes used in modern operating systems.
5	Describe file systems, disk scheduling, and device management techniques.
6	Apply operating system concepts to modern computing environments.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the structure and services of operating systems.	L2
2	Apply CPU scheduling algorithms for process management.	L3
3	Analyze synchronization and deadlock handling techniques.	L4
4	Demonstrate memory allocation and virtual memory management methods.	L3
5	Evaluate file system organization and disk scheduling techniques.	L5

6	Compare modern operating systems and emerging trends in OS design.	L4

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Computer system organization, interrupts, memory hierarchy, basics of system software, compiler, loader, linker, and process concepts		-
1	**Introduction to Operating Systems**	1.1Definition, objectives and functions of OS, evolution of operating systems, 1.2 types of operating systems: Batch, Multiprogramming, Multitasking, Time-sharing, Real-time, Distributed, Embedded and Mobile OS. 1.3OS structures: Monolithic, Layered, Microkernel. System calls, kernel and shell, OS services, booting process. **Self-learning Topics:** Linux vs Windows architecture, Open-source operating systems, Boot loaders, Mobile OS concepts. online reference: https://nptel.ac.in/courses/106106144 https://nptel.ac.in/courses/106102132	03	CO1
2	**Process Management and CPU Scheduling**	2.1Process concept, PCB, process states, process operations, threads and multithreading, inter-process communication, 2.2 CPU scheduling criteria, scheduling algorithms: FCFS, SJF, SRTF, Priority, Round Robin, 2.3 Multilevel Queue, Multilevel Feedback Queue scheduling. Numerical problems on scheduling algorithms. **Self-learning Topics:** Multithreading in modern applications, Context switching in Linux, Scheduling in multicore systems. online reference: https://nptel.ac.in/courses/106106144 https://archive.nptel.ac.in/content/storage2/courses/106104025	07	CO2
3	**Process Synchronization and Deadlocks**	3.1Critical section problem, synchronization hardware, semaphores, mutex locks, monitors, 3.2 classical synchronization problems (Producer-Consumer, Readers-Writers, Dining Philosophers), 3.3 deadlock characterization, prevention, avoidance, Banker's Algorithm, 3.4 deadlock detection and recovery. Numerical problems on Banker's Algorithm **Self-learning Topics:** Producer-consumer implementation in Linux, Deadlock handling in databases. online reference: https://nptel.ac.in/courses/106102132 https://www.coursera.org/learn/database-management	07	CO3

4	**Memory Management**	4.1Contiguous memory allocation, fragmentation, paging, segmentation, 4.2 virtual memory, demand paging, 4.3 page replacement algorithms: FIFO, LRU, Optimal, Thrashing, working set model, memory allocation policies. Numerical problems on paging and page replacement algorithms. **Self-learning Topics:** Virtual memory in modern OS, Cache memory optimization techniques. online reference: https://nptel.ac.in/courses/106105163	09	CO4
5	**File Systems and I/O Management**	5.1File concepts, file attributes, access methods, directory structures, 5.2 file allocation methods, free space management, 5.3 disk scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN, LOOK, 5.4 device management, RAID concepts, I/O hardware and software concepts. Numerical problems on disk scheduling algorithms. **Self-learning Topics:** NTFS vs EXT4 file systems, SSD storage management, Journaling file systems. online reference: https://nptel.ac.in/courses/106106144	10	CO5
6	**Modern Operating Systems, Protection and Security**	6.1 **OS Protection and Security:** Security goals, authentication, authorization, access control, 6.2 file protection, malware, firewall, intrusion detection, secure boot, 6.3mobile OS security, cloud security basics. 6.4 Study of latest operating systems: Windows 11, Ubuntu, Linux, Android, iOS, MacOS, Linux Mint **Self-learning Topics:** Zero Trust Security, Container Security, Cybersecurity in Cloud Platforms, Mobile Application Security online reference: https://nptel.ac.in/courses/106105217 https://nptel.ac.in/courses/106105167	09	CO6
		Total Hours	**45**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. A. Silberschatz, P. Galvin, G. Gagne, “Operating System Concepts”, 10th ed., Wiley, 2018.
2. W. Stallings, “Operating Systems: Internal and Design Principles”, 9th ed., Pearson, 2018.
3. A. Tanenbaum, “Modern Operating Systems”, Pearson, 4th ed., 2015.
4. Achyut Godbole and Atul Kahate, “Operating Systems”, McGraw Hill Education, 3rdEdition

5. N. Chauhan, “Principles of Operating Systems, 1st ed.”, Oxford University Press, 2014.
6. A. Tanenbaum and A. Woodhull, “Operating System Design and Implementation”, 3rd ed., Pearson.
7. R. Arpaci-Dusseau and A. Arpaci-Dusseau, ‘Operating Systems: Three Easy Pieces, CreateSpace Independent Publishing Platform”, 1st ed., 2018.

Online References:

Sr. No.	Website Name	URL
1	Swayam	https://nptel.ac.in/courses/106106144
2	NPTEL	https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_cs101
3	coursera	https://www.coursera.org/learn/operating-system-foundations

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a total of **six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	NL	
2024103	Computer Organization &Architecture	03	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024103	Computer Organization &Architecture	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Computer Organization and Architecture (COA) introduce the fundamental concepts of processor organization, instruction set architecture, memory hierarchy, and parallel processing. The course helps students understand the interaction between hardware and software for designing efficient computing systems.

Aligned with the theme “Arambh: From Prototype to Product”, it provides the foundation for developing and optimizing technology products through efficient hardware-software integration. In line with NEP 2020, the course promotes problem-solving, innovation, and industry readiness.

Course Objectives:

Sr. No.	Course Objectives
1	To understand the fundamentals of computer organization, architecture, functional units, and data transfer mechanisms in digital computer systems.
2	To analyze various number systems, data representation methods, and arithmetic algorithms used in computer systems.
3	To study processor organization, control unit design, instruction execution, memory interfacing, and assembly language programming concepts.
4	To understand memory organization, cache mechanisms, virtual memory, and input/output interfacing techniques in modern computer systems.
5	To explore parallel processing concepts, GPU architecture, CUDA architecture, and applications of GPUs in scientific and AI computing.
6	To examine advanced processor architectures, multicore systems, speculative execution techniques, and modern processor performance enhancement methods.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom’s Taxonomy
1	Explain the organization of computer systems, functional units, instruction cycle, and bus structures used in digital computers.	L2
2	Apply data representation techniques and arithmetic algorithms for performing binary computations and floating-point operations.	L3
3	Analyze processor architecture, control unit operations, addressing modes, interrupts, and memory interfacing techniques.	L4
4	Evaluate memory hierarchy, cache organization, virtual memory, and I/O data transfer mechanisms in computer systems.	L5
5	Illustrate GPU architecture, pipeline processing, SIMD concepts, and parallel computing techniques for high-performance applications.	L2

6	Examine advanced processor architectures and performance enhancement techniques.	L6

Detailed Syllabus:

Sr. No	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Number Systems, Boolean Algebra, Logic Gates, Binary Arithmetic, Fundamentals of Digital Electronics and Computer Systems.	02	-
1	**Overview of Computer Organization and Architecture**	1.1 Introduction to computer organization and architecture, Functional units of a digital computer, Von Neumann architecture. 1.2 Instruction cycle and CPU performance measures, Bus structures and data transfer mechanisms. **Self-learning Topics:** Gray Code, Error Detection Codes, ASCII and Unicode Representation. Online reference: https://nptel.ac.in/courses/106103068	06	CO1
2	**Data Representation and Arithmetic Algorithms**	2.1 Fixed-point and floating-point representation, IEEE 754 floating-point standard. 2.2 Binary Arithmetic, Booth's Algorithm, Restoring and Non-restoring Division Algorithms. **Self-learning Topics:** Carry Look Ahead Adder, Carry save adder, Wallace Tree Multiplication. Online reference: https://nptel.ac.in/courses/106106166	08	CO2
3	**Processor & Control Unit Design**	3.1 CPU Structure and Register Organization, Hardwired and Micro programmed Control Unit, 8086 Architecture, and Addressing Modes. 3.2 RISC and CISC Architecture, Interrupts and interrupt handling Memory interfacing. **Self-learning topic:** 8086 Assembly Language Programming. Online reference: https://microcontrollerslab.com/page/58/ https://microcontrollerslab.com/8086-logical-instructions-assembly-examples/ https://www.tutorialspoint.com/assembly_programming/assembly_tutorial.pdf	08	CO3
4	**Memory Organization and I/O Systems**	4.1 Memory hierarchy and cache organization, Cache mapping techniques, Cache coherence and associative memory. 4.2 Virtual memory, DMA and interrupt-driven I/O, I/O interfaces and controllers. **Self-learning topics:** Paging and segmentation, MESI Protocol Online reference:	08	CO4

		https://www.geeksforgeeks.org/operating-systems/difference-between-paging-and-segmentation/ https://www.geeksforgeeks.org/computer-organization-architecture/cache-coherence-protocols-in-multiprocessor-system/		
5	**GPU Architecture and Parallel Processing**	5.1 Pipeline processing and hazards, SIMD and parallel processing concepts, Introduction to GPU computing. 5.2 GPU architecture and CUDA architecture, Applications of GPUs in AI and scientific computing. **Self-learning topics:** Heterogeneous computing. Online reference: https://www.arm.com/glossary/heterogeneous-compute	08	CO5
6	**Advanced Processor Architectures**	6.1 Superscalar and multicore processors, Hyper-threading and speculative execution. 6.2 Advanced processors: Pentium, ARM and modern CPUs Performance enhancement techniques. **Self-learning topics:** Cluster computing Online reference: https://www.geeksforgeeks.org/computer-networks/an-overview-of-cluster-computing/	05	CO6
		Total Hours	**45**	

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References:

1. R. P. Jain, “Modern Digital Electronics”, Tata McGraw-Hill, 4th Edition / 2009.
2. Carl Hamacher and Zvonko Vranesic, “Computer Organization”, Tata McGraw-Hill, 5th Edition / 2002.
3. John P. Hayes, “Computer Architecture and Organization”, Tata McGraw-Hill, 3rd Edition / 2012.
4. William Stallings, “Computer Organization and Architecture: Designing for Performance”, Pearson Education, 8th Edition / 2010.
5. Douglas V. Hall, “Microprocessors and Interfacing”, Tata McGraw-Hill, 3rd Edition / 2017.
6. John Paul Shen and Mikko H. Lipasti, “Modern Processor Design: Fundamentals of Superscalar Processors, Waveland Press, 1st Edition / 2013.
7. Andrew S. Tanenbaum, “Structured Computer Organization”, Pearson Education, 6th Edition / 2012.
8. Kai Hwang, “Advanced Computer Architecture: Parallelism, Scalability, Programmability”, Tata McGraw-Hill, 3rd Edition / 2017.
9. Carl Hamacher, Zvonko Vranesic, and Safwat Zaky, “Computer Organization”, McGraw Hill Education, 6th Edition / 2011.
10. David A. Patterson and John L. Hennessy, “Computer Organization and Design: The Hardware/Software Interface”, Elsevier Inc., 5th Edition / 2012.

Online References:

Sr. No.	Website Name	URL
1	Nptel.Ac.In	https://nptel.ac.in/courses/106103068
2	Coursera.Org	https://www.coursera.org/learn/comparch
3	Edx.Org	https://www.edx.org/learn/computer-architecture
4	Nvidia	https://developer.nvidia.com/cuda-zone

Assessment:

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End Semester Theory Examination:

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- A total of **four questions** need to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2024105	Operating System Lab	--	--	02	--	--	30		30	1

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024105	Operating System Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The Operating System Laboratory provides practical exposure to Linux environment, process management, synchronization, memory management, file systems, disk scheduling, and system programming through hands-on experimentation. It enables students to understand, implement, and optimize core operating system concepts essential for developing reliable computing solutions. Aligned with the semester theme "Arambh: From Prototype to Product", the lab supports the transformation of software prototypes into efficient and scalable products through system-level design, testing, and performance optimization. The course also contributes to selected United Nations Sustainable Development Goals by promoting innovation, digital infrastructure, and quality technical education, particularly Goal 4: Quality Education, Goal 8: Decent Work and Economic Growth, and Goal 9: Industry, Innovation, and Infrastructure. The lab enhances problem-solving, technical competency, and industry readiness, supporting innovation-driven and sustainable technology development in alignment with NEP 2020.

Lab Objectives:

Sr. No.	Lab Objectives
1	Understand basic Linux operating system commands and environment.
2	Apply process scheduling algorithms for CPU management.
3	Implement process synchronization and deadlock handling algorithms.
4	Analyze memory management and page replacement techniques.
5	Demonstrate file management and disk scheduling algorithms.
6	Develop practical skills in operating system programming using Linux system calls and shell Scripting.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Execute Linux commands and shell programs in the operating system environment.	L3
2	Implement CPU scheduling algorithms and analyse their performance.	L3
3	Apply synchronization and deadlock handling techniques in operating systems.	L3

4	Demonstrate paging and memory management algorithms.	L3
5	Evaluate disk scheduling and file management methods.	L4
6	Develop system-level programs using operating system concepts and Linux utilities.	L4

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
1. Computer System with Intel i5 or higher processor 2. Minimum 8 GB RAM and 100 GB Storage 3. Keyboard, Mouse, and Internet Connectivity	1. Windows/Linux Operating System 2. GCC Compiler / Code::Blocks / VS Code 3. Web Browser for Online Learning Resources

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	**Prerequisite**	Linux environment, terminal commands, gcc compiler, Ubuntu installation basics. (Prerequisite syllabus should not be considered for paper setting)	02	-
1	**Linux Commands and Shell Programming**	1.1 Linux file and directory commands, file permissions, process-related commands, shell scripting basics, command-line utilities. **Self-learning Topics**: Bash shell customization, Linux user administration. Online reference: https://nptel.ac.in/courses/106106144	05	LO1
2	**CPU Scheduling Algorithms**	2.1 Implementation and analysis of non-pre-emptive FCFS, SJF, scheduling algorithms 2.2 Implementation and analysis of pre-emptive Priority Scheduling, Round Robin, Multilevel Queue scheduling algorithms. **Self-learning Topics:** Scheduling in multicore processors. Online reference: https://nptel.ac.in/courses/106106144	05	LO2
3	**Synchronization and Deadlock**	3.1 Implementation of semaphores, Producer-Consumer problem, Reader-Writer problem, Dining Philosopher problem, 3.2 Deadlock avoidance Banker's Algorithm. **Self-learning Topics:** Mutex and synchronization in Linux. Online reference: https://nptel.ac.in/courses/106106144	05	LO3
4	**Memory Management**	4.1 Paging, segmentation, memory allocation techniques. 4.2 FIFO, LRU, Optimal Page Replacement algorithms, **Self-learning Topics:** Virtual memory implementation in Linux.	06	LO4

		Online reference: https://nptel.ac.in/courses/106106144		
5	**File System and Disk Scheduling**	5.1 File handling system calls, file permissions, 5.2 disk scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN). **Self-learning Topics:** Linux file system architecture (EXT4). Online reference: https://nptel.ac.in/courses/106106144	05	LO5
6	**System Programming and Advanced Concepts**	6.1 fork(), exec(), wait(), pipe(), shared memory, process communication, 6.2 basic system monitoring utilities. **Self-learning Topics:** Containers and virtualization basics. Online reference: https://nptel.ac.in/courses/106106144	04	LO6
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination**.**				

References:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, “Operating System Concepts”, Wiley Publication
2. Andrew S. Tanenbaum (later with Herbert Bos), “Modern Operating Systems”, 5th edition,Prentice Hall / Pearson
3. William Stallings, “Operating Systems: Internals and Design Principles”,9th edition,Pearson.
4. Robert Love, O’reilly publications, “Linux System Programming: Talking Directly to the Kernel and C Library”, Second Edition.

Online References:

Sr. No.	Website Name	URL
1	SWAYAM	https://nptel.ac.in/courses/106106144
2	NPTEL	https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_cs101

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	Study and execution of basic Linux commands	2
02	Linux file handling and directory management commands	2
03	Shell scripting for arithmetic operations and conditional statements	2
04	Shell scripting for file handling and process management	2
05	Implementation of FCFS CPU Scheduling Algorithm	2
06	Implementation of SJF and Priority Scheduling Algorithms	2
07	Implementation of Round Robin CPU Scheduling Algorithm	2
08	Simulation of Producer–Consumer Problem using Semaphores	2

09	Implementation of Reader–Writer Problem	2
10	Implementation of Banker's Algorithm for Deadlock Avoidance	2
11	Implementation of FIFO Page Replacement Algorithm	2
12	Implementation of LRU and Optimal Page Replacement Algorithms	2
13	Implementation of Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN)	2
14	Study of Process Creation using fork(), exec(), wait() System Calls	2
15	Inter Process Communication using Pipe and Shared Memory	2

Suggested List of Assignment:

Sr No	List of Assignments / Tutorials	Hours
01	Comparative study of Linux and Windows Operating Systems	2
02	Numerical problems on CPU Scheduling Algorithms	2
03	Numerical problems on Banker's Algorithm	2
04	Comparative analysis of Page Replacement Algorithms	2
05	Study of latest operating systems: Windows 11, Ubuntu Linux, Android, iOS	2
06	Study of virtualization and containers	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2024106	Nirman Lab-IV	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024106	Nirman Lab-IV	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Aligned with the National Education Policy (NEP) 2020, the institution emphasizes experiential, interdisciplinary, and project-based learning through the Nirman Lab a central hub for hands-on innovation. To strengthen the undergraduate research ecosystem, the institution has adopted a theme-based academic model aligned with UNSGD. Each semester features six curated problem statements based on local need and aligned with core subjects in the same semester, enabling students to apply classroom knowledge to real-world challenges. Every student selects one problem and develops an individual, subject-integrated solution enhancing both academic understanding and research skills. The Nirman Lab supports this initiative with facilities for design thinking, prototyping, and product development. Students maintain a project logbook throughout the semester to track their progress and reflections

Lab Objectives:

Sr. No.	Lab Objectives
1	To promote experiential and project-based learning that bridges theoretical knowledge with real-world problem-solving.
2	To encourage interdisciplinary integration by enabling students to apply concepts from multiple subjects within a single cohesive project.
3	To develop innovation and design thinking skills through hands-on activities and iterative solution development.
4	To foster critical thinking and creativity by engaging students in open-ended problems with multiple solution pathways.
5	To enhance communication, collaboration, and documentation skills essential for professional engineering practice.
6	To build an entrepreneurial and research mindset by guiding students to develop scalable socially-relevant, and technically viable prototype.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Recall and articulate key concepts from core and allied subjects relevant to the assigned project	L1

2	Explain the interdisciplinary nature of the problem and the role of each subject in addressing it	L2
3	Apply appropriate tools, techniques, and theoretical knowledge to develop project components	L3
4	Analyze problem constraints and user requirements to structure a feasible and efficient solution	L4
5	Evaluate multiple design options and justify the chosen solution based on technical and practical considerations	L5
6	Create a functional prototype or solution that demonstrates innovation, utility, and integration of interdisciplinary knowledge	L6

Project Guidelines (Interdisciplinary Project Execution in Nirman Lab)

- Each student works on an individual interdisciplinary project aligned with the semester theme.
- Faculty in-charges mentor both the academic and technical aspects, and track weekly progress.
- Project assessment will be rubric-based, ensuring depth, innovation, documentation, and ownership.
- Students shall convert the best solution into working model using various components of their domain areas and demonstrate.

Guidelines for same semester Subject Concepts Applied within the Project

- Term work for each subject will partially reflect how well a student applies subject-specific concepts in their interdisciplinary project.
- Internal assessment panel will collaborate to align project components with subject learning outcomes.

Role of Faculty In-Charges in Nirman Lab Projects

Faculty in-charges play a pivotal role in the success of interdisciplinary, theme-based projects under the Nirman Lab. Their responsibilities extend beyond technical supervision to include academic alignment, innovation facilitation, and active student engagement.

Their key roles include:

- Motivating and Inspiring Students
  - Encourage students to take ownership of their learning and projects.
  - Cultivate a mindset of curiosity, exploration, and social relevance.
  - Foster an environment where students feel empowered to take creative risks.
- Conducting Brainstorming and Ideation Sessions
  - Organize structured brainstorming sessions at the start of the semester to help students define their problem statements and solution pathways.
  - Promote collaborative thinking, design exploration, and interdisciplinary integration.
- Arranging Guest Lectures and Expert Talks
  - Identify and invite industry experts, researchers, and innovators for guest lectures aligned with the semester's theme or subject areas.
  - Facilitate exposure to real-world challenges, current trends, and future opportunities.
- Ensuring Uniqueness and Originality of Projects
  - Actively review proposed ideas to ensure **no duplication of solutions** across students.
  - Encourage students to explore novel approaches, technologies, and perspectives.
- Promoting Discussion and Collaborative Learning
  - Create platforms for students to present, discuss, and receive peer and mentor feedback.
  - Facilitate idea refinement through regular discussions and group engagement.
- Monitoring and Documentation
  - Oversee project logbook maintenance, milestone tracking, and submission of progress reports.
  - Provide ongoing feedback and ensure project alignment with learning outcomes.
- Coordination with Subject Faculty

- Work in collaboration with other subject faculty to help students embed theoretical and practical aspects of their coursework into the project.
- Facilitate subject-term mapping and contribute to termwork assessment based on evidence.

Guidelines for Assessment

Presentation-Based Assessment Structure (Total: 25 Marks)

Assessment Month	Weightage	Marks
Month 1 (Formative 1)	20%	5 marks
Month 2 (Formative 2)	40%	10 marks
Month 3 (Formative 3)	40%	10 marks

Rubric-Based Evaluation Criteria

Criteria	Month 1 (5)	Month 2 (10)	Month 3 (10)
Problem Understanding	Connects problem to subjects	Defines interdisciplinary scope	Demonstrates deep conceptual grasp
Subject Knowledge Application	Identifies relevant concepts	Applies principles in design	Integrates multiple subject areas correctly
Innovation & Design Thinking	Proposes creative idea	Develops and tests feasible solution	Final solution shows originality and utility
Documentation & Presentation	Logbook initiated, plan presented	Mid-design log and visuals	Final report and demo completed
Progress & Ownership	Meets deadlines, shows planning	Demonstrates self-motivation	Completes project independently with reflection

Total Assessment Structure

Component	Marks	Assessed By
Termwork – Project Execution	25 Marks	Project Guide
Viva Voce (Final Evaluation)	25 Marks	External Examiner

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2024401	Linux Operating System	--	--	2*+2	--	--	60		60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2024401	Linux Operating System	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The Linux Operating System course provides hands-on knowledge of Linux fundamentals, system administration, shell scripting, networking, and automation. It enables students to develop, manage, and deploy software solutions using open-source technologies, supporting the transition from prototype to product while enhancing industry readiness and problem-solving skills. UNSDG Alignment: SDG 4 (Quality Education), SDG 8 (Decent Work & Economic Growth), and SDG 9 (Industry, Innovation & Infrastructure).

Lab Objectives:

Sr. No.	Lab Objectives
1	Understand Linux architecture and open-source ecosystem.
2	Use Linux commands and utilities for system operation.
3	Develop shell scripts for automation and product prototyping.
4	Manage files, users, processes, and permissions securely.
5	Configure networking and deployment-related services in Linux.
6	Apply Linux tools in software product development environments.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain Linux architecture, kernel, shell, and file system concepts.	L2
2	Apply Linux commands and utilities for efficient system operations.	L3
3	Develop shell scripts for automation and prototype development.	L3
4	Analyse process management and Linux security mechanisms.	L4
5	Configure Linux networking and package management tools.	L4
6	Build Linux-based deployment environments for software products.	L5

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Intel Core i3 / AMD Ryzen 3 Processor or Higher Minimum 4 GB RAM (8 GB Recommended) Minimum 50 GB Free Disk Space 64-bit Architecture Support	Linux Operating System (Ubuntu/Fedora/Debian) VirtualBox or VMware Workstation Player (Optional) Text Editor (Vim, Nano, or Visual Studio Code) Web Browser (Firefox, Chrome, etc.)

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	**Prerequisite**	CLI basics, Operating System concepts, file handling.	04	-
1	**Introduction to Linux**	1.1 Linux history, GNU project ,Linux architecture, kernel, shell 1.2 Distributions, installation basics 1.3 Open-source licensing, theme alignment: Linux as product innovation platform. **Self-Learning Topics:** Comparison of popular Linux distributions (Ubuntu, Fedora, Debian, CentOS), Linux usage in Cloud Computing, DevOps, and Embedded Systems, Study of open-source licenses: GPL, MIT, Apache License. Online reference: https://ubuntu.com, https://fedoraproject.org, https://www.debian.org, https://www.gnu.org/licenses, https://opensource.org/licenses.	4	LO1
2	**Linux Commands and File System**	2.1 Linux file hierarchy, commands 2.2 File handling, filters, pipes 2.3 Redirection, editors, permissions and ownership. **Self-Learning Topics:** Advanced usage of grep, awk, and sed commands, Comparison of text editors: vi, nano, and gedit, File compression and archiving utilities using tar, gzip, and zip. Online reference: https://www.gnu.org/software/grep/manual/ https://linuxjourney.com	12	2
3	**Shell Programming and Automation**	3.1 Shell scripting, variables, operators 3.2 oops, conditions, functions 3.3 Automation scripts. **Self-Learning Topics:** Cron jobs and task scheduling in Linux, Environment variables and shell customization, Basics of shell script debugging and optimization Online reference: https://www.coursera.org/learn/linux-commands-and-shell-scripting-bits	12	LO3

		https://www.coursera.org/learn/linux-shell-scripting-build-automate-troubleshoot		
4	**Process and User Management**	4.1 Process lifecycle, scheduling, process monitoring, User/group management 4.2 Linux security concepts **Self-Learning Topics:** System log analysis and monitoring using log files in /var/log, Configuration and usage of sudo and administrative privileges in Linux, Introduction to SELinux /AppArmor security mechanisms in Linux systems. Online reference: https://linuxjourney.com https://www.digitalocean.com/community/tutorials https://www.redhat.com/en/topics/linux/what-is-selinux?	12	LO4
5	**Linux Networking and Package Management**	5.1 IP configuration, SSH, SCP 5.2 Package management, remote administration, services. **Self-Learning Topics:** Firewall configuration basics using ufw and iptables, Linux networking in cloud and virtualized environments, Secure Shell (SSH) key-based authentication and remote server administration. Online reference: https://www.digitalocean.com/community/tutorials https://docs.aws.amazon.com https://www.openssh.com/manual.html	12	LO5
6	**Linux for Product Development**	6.1 Linux in cloud, containers 6.2 Git basics, DevOps overview 6.3 Deployment lifecycle and case studies. **Self-Learning Topics:** Advanced Docker Commands and Docker Compose, Linux-based Cloud Platforms (AWS EC2, Azure VM, Google Cloud VM), Linux Server Hardening Techniques. DevOps Lifecycle and Industry Practices Online reference: https://docs.docker.com https://aws.amazon.com/training/digital/aws-cloud-practitioner-essentials/ https://www.atlassian.com/devops	4	LO6
Total			**60**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Harwani, B. M. (2015),” UNIX and Shell Programming (1st ed.)”, Oxford University Press, India.

2. Soyinka, Wale. (2016), “Linux Administration: A Beginner's Guide (7th ed.)”, McGraw-Hill Education.
3. Shotts, William E. Jr. (2019),”The Linux Command Line: A Complete Introduction (2nd ed.)”, No Starch Press.
4. Ward, Brian. (2021), “How Linux Works: What Every Superuser Should Know (3rd ed.)”, No Starch Press.

Online References:

Sr. No.	Website Name	URL
1	Linux Guide	https://tldp.org
2	GNU Bash Ref	https://www.gnu.org/software/bash/manual/bash.html
3	Linuxjourney	https://linuxjourney.com

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	Installation of Linux Operating System using Virtual Machine: Installation of Ubuntu/Linux distribution using VirtualBox or VMware, Understanding partitioning and boot process.	4
02	Write shell scripts using variables and operators to perform arithmetic operations and user input handling.	4
03	Shell scripting using variables and operators: Programs for menu-driven applications, factorial, palindrome, and file operations.	4
04	Create automation scripts using functions and command-line arguments: Automate backup, log generation, or file monitoring tasks.	4
05	Study and implementation of process monitoring commands using ps, top, kill, and nice.	4
06	Creation and management of users and groups using Linux administrative commands.	4
07	Implementation of Linux file permissions and security policies using chmod, chown, and umask.	4
08	Configure IP address and verify network connectivity using ping, ifconfig, and ip commands.	4
09	Perform secure remote login and file transfer using SSH and SCP.	4
10	Install, update, and manage software packages and services using apt/yum and systemctl.	4
11	Network configuration using Linux utilities	4
12	Remote login and file transfer using SSH/SCP	4
13	Package installation and service management	4

Suggested List of Assignments:

Sr No	List of Assignments / Tutorials	Hours
01	Linux Fundamentals and System Administration Compare Ubuntu, Fedora, Debian, and CentOS based on architecture, package management, release cycle, and applications. Demonstrate Linux file management using commands, pipes, filters, redirection, and file permissions. Create users and groups, assign permissions, and monitor running processes using Linux administrative commands. Prepare a report explaining the role of Linux in Cloud Computing, DevOps, and Embedded Systems.	2

02	Shell Scripting and Linux Security Write shell scripts using variables, operators, loops, conditional statements, and functions to automate routine system tasks. Analyze Linux log files located in /var/log and identify security-related events. Configure sudo privileges for different users and explain the principle of least privilege. Study SELinux/AppArmor and prepare a comparative report on their security features and use cases.	2
03	Linux Networking, Cloud, and DevOps Configure basic firewall rules using ufw or iptables and document the results. Set up SSH key-based authentication for secure remote login and perform basic remote administration tasks. Create and manage containers using Docker and Docker Compose for a multi-service application. Prepare a case study on Linux-based cloud platforms (AWS EC2, Azure VM, Google Cloud VM), Linux server hardening techniques, and the DevOps lifecycle followed in industry.	2

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2994501	Project Management	--	--	2*+2	--	--	60	--	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2994501	Project Management	---	---	---	---	---	50	--	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Project Management Lab provides hands-on learning in project planning, scheduling, cost estimation, risk management, resource allocation, quality assurance, and project monitoring for engineering and IT projects. Aligned with NEP 2020, it promotes experiential learning, critical thinking, teamwork, problem-solving, and industry-relevant skills. The course also supports UNSDG 4 (Quality Education), UNSDG 8 (Decent Work and Economic Growth), UNSDG 9 (Industry, Innovation and Infrastructure), and UNSDG 12 (Responsible Consumption and Production) by fostering sustainable project management practices, innovation, and efficient resource utilization.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand project management concepts, project life cycle, and PMBOK framework.
2	To apply project initiation, feasibility analysis, and project charter preparation techniques.
3	To develop project scope, Work Breakdown Structure (WBS), and project schedules.
4	To perform project cost estimation, budgeting, and risk management activities.
5	To apply stakeholder, communication, resource, and quality management practices.
6	To evaluate project execution, monitoring, control, and project closure activities.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain project management concepts and organizational structures.	L2
2	Apply project initiation and integration management techniques.	L3
3	Develop WBS, CPM/PERT schedules, and project plans.	L3
4	Analyze project costs, budgets, risks, and quality parameters.	L4
5	Apply stakeholder, communication, and resource management practices.	L3
6	Evaluate project monitoring, control, and closure activities.	L5

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Computer/Laptop with Internet Connectivity Projector / Smart Classroom	MS Project / OpenProject / ProjectLibre Microsoft Excel Draw.io / Lucidchart MS Office / LibreOffice Web Browser

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	**Prerequisite**	Basic understanding of engineering projects, software development process and management principles.	02	-
1	**Project Management Foundations**	Project Management Foundations: 1. Project, Project Characteristics, Project Life Cycle, Stakeholders, Triple Constraints, 2. PMBOK Framework, Organizational Structures, Role of Project Manager. **Self-Learning Topics:** Agile and Hybrid Project Management. Online Reference: https://nptel.ac.in/courses/110104073	08	LO1
2	**Project Initiation and Integration Management**	Project Initiation and Integration Management: 2.1 Project Selection Methods, Feasibility Analysis, Business Case, Project Charter, 2.2 Project Integration Management, Project Governance and Project Planning. **Self-Learning Topics:** Startup Project Evaluation and Product Feasibility. Online Reference: https://nptel.ac.in/courses/110104073	10	LO2
3	**Scope and Time Management**	Scope and Time Management: 3.1 Scope Planning, Scope Definition, Work Breakdown Structure (WBS), Scope Verification, Scope Change Control, 3.2 Activity Definition, Activity Sequencing, Network Diagrams, CPM and PERT. **Self-Learning Topics:** Project Scheduling Tools. Online Reference: https://www.coursera.org/learn/project-scheduling	10	LO3
4	**Cost and Risk Management**	Cost and Risk Management: 4.1 Project Cost Estimation, Budgeting, Cost Control, Earned Value Management (EVM), 4.2 Risk Identification, Risk Analysis, Risk Response Planning and Monitoring. **Self-Learning Topics:** Monte Carlo Risk Analysis.	10	LO4

		Online Reference: https://www.coursera.org/learn/project-risk-management		
5	**Resource, Communication and Quality Management**	Resource, Communication and Quality Management: 5.1 Human Resource Management, Team Building, Leadership, Communication Planning, 5.2 Stakeholder Management, Quality Planning, Quality Assurance and Quality Control. **Self-Learning Topics:** Virtual Team Management. Online Reference: https://www.coursera.org/learn/project-management-communication-stakeholders-and-leadership	10	LO5
6	**Project Execution, Monitoring and Closure**	Project Execution, Monitoring and Closure: 6.1 Project Execution, Monitoring and Control, Change Management, Procurement Management, 6.2 Contract Management, Project Audit, Project Closure and Lessons Learned. **Self-Learning Topics:** Product Deployment and Post-Implementation Review. Online Reference: https://www.coursera.org/learn/project-execution-google	10	LO6
	Total		**60**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Harold Kerzner, “Project Management: A Systems Approach to Planning, Scheduling and Controlling”, Wiley, 13th Edition / 2022.
2. Kathy Schwalbe, “Information Technology Project Management”, Cengage Learning, 10th Edition / 2021.
3. Clifford F. Gray and Erik W. Larson, “Project Management: The Managerial Process”, McGraw-Hill Education, 8th Edition / 2020.
4. Prasanna Chandra, “Projects: Planning, Analysis, Selection, Financing, Implementation and Review”, McGraw-Hill Education, 10th Edition / 2023.
5. Project Management Institute (PMI), “A Guide to the Project Management Body of Knowledge” ,Project Management Institute, 7th Edition / 2021.

Online References:

Sr. No.	Website Name	URL
1	NPTEL Project Management	https://nptel.ac.in/courses/11010407

2	PMI	https://www.pmi.org/
3	Coursera Project Management	https://www.coursera.org/professional-certificates/google-project-management

Suggested List of Experiments.

Sr No	List of Experiments	Hours
01	Project Life Cycle Analysis and Stakeholder Identification	02
02	Project Charter Preparation for an Engineering/IT Project	02
03	Feasibility Analysis and Project Selection Techniques	02
04	Work Breakdown Structure (WBS) Development	02
05	Project Scheduling using CPM and PERT	02
06	Project Cost Estimation and Budget Preparation	02
07	Risk Identification, Risk Register and Risk Response Planning	02
08	Stakeholder Analysis and Communication Management Plan	02
09	Resource Allocation Matrix and Quality Management Plan	02
10	Project Monitoring, Change Control and Project Closure Report	02
11	Development of Project Governance Structure and Responsibility Assignment Matrix (RAM/RACI)	02
12	Preparation of Gantt Chart and Milestone Tracking Plan Using Project Management Tools	02
13	Earned Value Management (EVM) Analysis for Project Performance Evaluation	02
14	Procurement Planning and Contract Management for Engineering Projects	02
15	Project Audit, Lessons Learned Documentation and Post-Implementation Review.	02

Suggested List of Assignments:

Sr No	List of Assignments / Tutorials	Hours
01	Assignment 1 (Modules 1–3): Prepare Project Charter, WBS and Project Schedule for a selected Engineering/Software Project.	02
02	Assignment 2 (Modules 4–6): Perform Cost-Risk Analysis, Stakeholder Management Planning and Project Closure Report for a selected Project.	02

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 50 Marks (Total) = 25 Marks (Experiments) + 10 Marks (Assignments) + 05 Marks (Attendance) + 10 (Presentation)

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2994502	Design Thinking	--	--	2*+2	--	--	60	--	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2994502	Design Thinking	---	---	---	---	---	50	--	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Modern computer engineers must develop innovative, user-centric solutions to address real-world challenges. This course introduces Design Thinking as a human-centered approach to creativity, problem-solving, prototyping, and product development. Aligned with the theme, NEP 2020 and with UNSDG goals of SDG 4,8,9 and SDG 11 in such a way that it enables students to transform ideas into impactful digital solutions while fostering innovation, critical thinking, collaboration, and entrepreneurship.

Lab Objectives:

Sr. No.	Lab Objectives
1	To introduce a learner to the principles of Design Thinking.
2	To familiarize a learner with the process (stages) of Design Thinking.
3	To introduce various design thinking tools.
4	Study of the techniques for generation of solutions for a problem.
5	To expose a learner to various case studies of Design Thinking.
6	Create and test a prototype.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Compare traditional approach to problem solving with the Design Thinking approach and discuss the principles of Design Thinking	L1
2	Define a user persona using empathy techniques	L2
3	Frame a problem statement using various Design Thinking tools	L3
4	Use ideation techniques to generate a pool of solutions for a problem	L4
5	Create prototypes using different techniques	L5
6	Test the prototypes and gather feedback for refining the prototype	L6

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:

Computer/Laptop with Internet connectivity Projector/Smart Classroom Basic prototyping materials	Operating System (Windows/Linux/macOS) Web Browser Office Suite (MS Office/LibreOffice) Design & Collaboration Tools (Figma/Canva/Miro or equivalent)

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	Prerequisite	Basic understanding of problem-solving approaches, Communication and presentation skills, Creativity and analytical thinking, Team collaboration skills.	02	-
1	**Introduction to Design Thinking**	1.1 Definition, Comparison of Design Thinking and traditional problem-solving approach, Need for Design Thinking approach, Key tenets of Design Thinking 1.2 Five Stages of Design Thinking (Empathize, Define, Ideate, Prototype, Test). **Self-Learning Topics:** Analysis of Design Thinking case studies in Healthcare, Education, Smart Cities, FinTech, Sustainability and Digital Product Development. Online Reference: https://onlinecourses.swayam2.ac.in/aic23_ge17/preview	10	LO1, LO2
2	**Empathy**	2.1 Foundation of empathy, Purpose of empathy, Observation for empathy, 2.2 User observation technique, Creation of empathy map. **Self-learning Topics:** Develop Empathy Maps and User Personas using real-world user observations, interviews and surveys. Online Reference: https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_mg51 https://www.coursera.org/learn/user-research	10	LO2, LO3
3	**Define**	3.1 Significance of defining a problem, Rules of prioritizing problem solving, 3.2 Conditions for robust problem framing, Problem statement and POV. **Self-learning Topics:** Problem Framing using Personas, Point-of-View (POV) Statements and User Need Analysis. Online Reference: https://www.coursera.org/learn/uva-darden-design-thinking-innovation	10	LO2, LO3
4	**Ideate**	4.1 What is ideation? Need, Ideation techniques, Guidelines for ideation: 4.2 Multi-disciplinary approach, imitating with grace, Breaking patterns, challenging assumptions, Looking across value chain, Looking beyond recommendation, 4.3 Techniques for ideation: Brainstorming, Mind mapping.	10	LO3

		Self-learning Topics: Advanced Ideation Techniques: SCAMPER, Brainwriting, Reverse Thinking, Crazy-8s and Innovation Workshops. Online Reference: https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_de15		
5	**Prototype**	5.1 Low and high-fidelity prototypes, Paper prototype, 5.2 Story board prototype, Scenario prototype. **Self-learning Topics:** Hands-on exploration of Figma, Adobe XD, Canva, Balsamiq and Rapid Prototyping for Digital Products. Online Reference: https://www.udemy.com/course/design-thinking-masterclass/ https://nptel.ac.in/courses/110106124	10	LO6
6	**Test:**	6.1 Guidelines of conducting test, The end goals of test: 6.2 Desirability, Feasibility and Viability, Usability testing **Self-learning Topics:** Usability Testing Strategies, A/B Testing, User Feedback Analysis and Iterative Design Improvement. Online Reference: https://www.coursera.org/learn/conduct-ux-research	08	LO4, LO5
Total			**60**	

Note:
1.Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2.Course Engagement (ARCS-based): The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Pavan Soni, “Design Your Thinking: The Mindsets, Toolsets, and Skill Sets for Creative Problem-Solving”, Penguin Random House India Private Limited, 2016.
2. Emrah Yayici, “Design Thinking: Methodology Book”, CRC Press, 1st Edition, 2016.
3. Christian Müller-Roterberg, “Handbook of Design Thinking, CreateSpace Independent Publishing Platform”, 1st Edition, 2018.
4. Idris Mootee, “Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School”, John Wiley & Sons, 1st Edition, 2013.
5. Tim Brown, “Change by Design: How Design Thinking Creates New Alternatives for Business and Society”, Harper Business, Revised Edition, 2009.

Online References:

Sr. No.	Website Name	URL
1	swayam.gov.in	Design Thinking and Innovation by Ravi Poovaiah https://onlinecourses.swayam2.ac.in/aic23_ge17/preview
2	Swyam.gov.in	Introduction to Design Thinking by Dr. Rajeshwari Patil, Dr. Manisha Shukla, Dr. Deepali Raheja, Dr. Mansi Kapoor https://onlinecourses.swayam2.ac.in/imb24_mg37/preview

3	Interaction-design.org	Usability Testing https://www.interaction-design.org/literature/topics/usability-testing

Suggested List of Experiments:

Sr No	List of Experiments	Hours
01	Customer Journey Mapping: Visualize the steps users take to interact with a product or service. Map out the customer journey from discovering a product to making a purchase and using the product. Identify pain points and opportunities for improvement.	02
02	Stakeholder mapping: Identify all relevant stakeholders in a project. Create a stakeholder map, categorizing stakeholders based on their influence and interest. Include management of relationships with key stakeholders.	02
03	"How Might We" Problem Framing: Transform user insights into actionable problem statements. After empathizing with users, turn challenges into "How Might We" statements that define the problem without prescribing a solution.	02
04	Brainstorming Session: Generate a pool of ideas in a creative, non-judgmental environment. Using ideation techniques like mind mapping and brainwriting, students brainstorm as many solutions as possible to their "How Might We" problem statements.	02
05	Affinity Diagramming: Organize group ideas to find patterns and insights. After brainstorming, students will categorize their ideas into themes by placing sticky notes on a wall and moving them into groups based on similarities.	02
06	Rapid Prototyping: Create quick, low-fidelity versions of solutions. Use materials like paper, cardboard, and markers to build a prototype of their solution within 30 minutes. The focus is on speed and functionality, not aesthetics.	02
07	Wireframing: Create a visual guide for digital interfaces for mobile app / web app for the problems identified in earlier lab sessions. Students will sketch wireframes of the user interface for their product or service. Use tools like Balsamiq or paper and pen for low-fidelity wireframes.	02
08	Role-Playing: Walk through a prototype from the user's perspective. Students act as both users and designers, role-playing scenarios where they interact with their prototype (Developed in earlier lab sessions). Gather feedback from participants on how to improve the experience.	02
09	Usability Testing: Evaluation of the effectiveness and user-friendliness of a prototype (developed in earlier lab sessions). Students will have peers or target users test their prototypes, observe how they interact with it, and collect feedback on any issues or improvements needed.	02
10	Feedback Loop and Iteration: Refine solutions based on user feedback. After usability testing, students will refine their prototypes. Document changes made based on feedback and discuss how continuous iteration improves the design.	02
11	Design Thinking Canvas: Develop a complete Design Thinking Canvas for a selected real-world problem.	02
12	Competitor Analysis and Benchmarking: Analyze existing solutions and identify innovation opportunities.	02
13	Crazy-8 Ideation Workshop: Generate multiple solution concepts using Crazy-8 technique.	02
14	Interactive Digital Prototype Development using Figma/Canva.	02
15	Prototype Pitch, Validation and Stakeholder Feedback Analysis.	02

Suggested List of Assignment:

Sr No	List of Assignments / Tutorials	Hours
01	Conduct user observation and prepare an Empathy Map and User Persona for a real-world problem.	02
02	Frame a Problem Statement (POV) and generate multiple solution ideas using Brainstorming and Mind Mapping techniques.	02
03	Develop a low-fidelity prototype, conduct usability testing, and document user feedback with suggested improvements.	02

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 50 Marks (Total) = 25 Marks (Experiments) + 10 Marks (Assignments) + 05 Marks (Attendance) + 10 (Presentation)

Justification for B.E. (Computer Engineering)

1.	Necessity for starting the course:	The demand for Computer Engineering professionals is consistently high, and individuals with a B.E. in Computer Engineering can find opportunities in various sectors, including core companies, IT services companies, Product-based companies, Startups, Govt. and more.
2.	Opportunities of Employability / Employment available after undertaking these courses:	B.E. in Computer Engineering can open up various opportunities and employment prospects across various private industries and government sectors. Additionally, as electronics continue to advance, new roles and specialties within Computer Engineering field are continually emerging, providing diverse career paths for B.E. Computer Engineering graduates. With various roles as Software Developer, Web Developer, Front-end Developer, Full-Stack Developer, Data Analyst, Security Analyst, Cloud Engineer, AI Engineer, Network Engineer, QA Engineer, Database Administrator and many more.

Dr. Kishor Sakure
BoS-Chairman –Computer Engineering

Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-Academic Council

Dr. Deven Shah
Principal
Chairman-Academic Council

Letter Grades and Grade Points:

Semester GPA/ Programme CGPA Semester/ Programme	% of Marks	Alpha-Sign/ Letter Grade Result	Grading Point
9.00 - 10.00	90.0 – 100	O (Outstanding)	10
8.00 - < 9.00	80.0 - < 90.0	A+ (Excellent)	9
7.00 - < 8.00	70.0 - < 80.0	A (Very Good)	8
6.00 - < 7.00	60.0 - < 70.0	B+ (Good)	7
5.50 - < 6.00	55.0 - < 60.0	B (Above Average)	6
5.00 - < 5.50	50.0 - < 55.0	C (Average)	5
4.00 - < 5.00	40.0 - < 50.0	P (Pass)	4
Below 4.00	Below 40.0	F (Fail)	0
Ab (Absent)	-	Ab (Absent)	0